

Autodesk® Scaleform®

Scaleform CLIK AS2 User Guide

CLIK

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: 2.0
: 2010 8 19

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Autodesk® Scaleform® 4.3

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1 소개

Common Lightweight Interface Kit (CLIK™)

[Getting Started with CLIK](#) [Getting Started with CLIK Buttons](#) 2

CLIK
CLIK

1.1

CLIK 2.0(AS2) , PC
Scaleform 4.1 (CPU) ,
UI
CLIK UI , 15
(, , ,)

1.1.1. 스케일폼 CLIK 에 포함된 것

컴포넌트: UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	ProgressBar
RadioButton	UILoader
Label	ScrollingList
TextInput	TileList
TextArea	DropDownMenu
ScrollIndicator	Dialog
ScrollBar	NumericStepper

클래스: API

InputManager	Locale
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
EventDispatcher	IList
Delegate	IListItemRenderer
Constraints	GameDataProvider

문서와 예제 파일

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK API Reference
CLIK User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2

gSkinner.com

CLIK

Grant Skinner	gskinner.com	CS4	
		Flash team	.
gskinner.com		http://gskinner.com/blog	.

CLIK

Flash Studio

CLIK

C:\Program Files\Scaleform\GFx 4.3\Resources\AS2\CLIK\components\CLIK_Components fla

1.3 UI

UI (Microsoft Visio®)
(visual diagram editor)

. CLIK

UI (Visio)
MovieClip

UI (workflow)

()

, UI

UI (layout)

UI

UI

(

)

2 프리빌트 컴포넌트(Prebuilt Components)

CLIK UI . , CLIK
() CLIK
.

CLIK UI .
.

.

- :
- :
- : ()
- :
- : UI
- :

2.1 (Basic Button and Text Types)

Button, CheckBox, Label, TextInput, TextArea . Button
, CheckBox Button
TextInput TextArea textField , Label .



그림 1: *Mass Effect* 의 Main Menu Button 예제

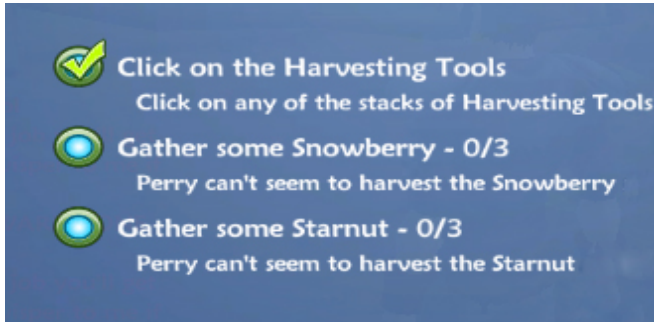


그림 2: *Free Realms* 의 CheckBox 예제



그림 3: *Free Realms* 의 Label 예제



그림 4: *Crysis Warhead* 의 TextInput 예제

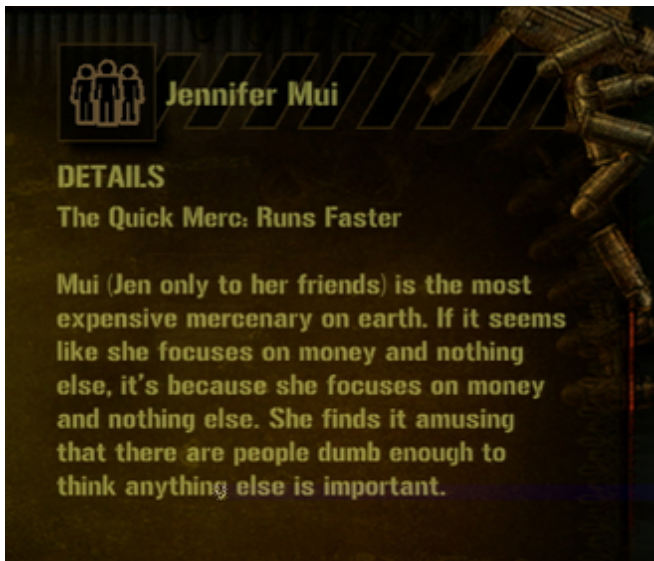


그림 5: *Mercenaries 2* 의 TextArea 예제

2.1.1 Button

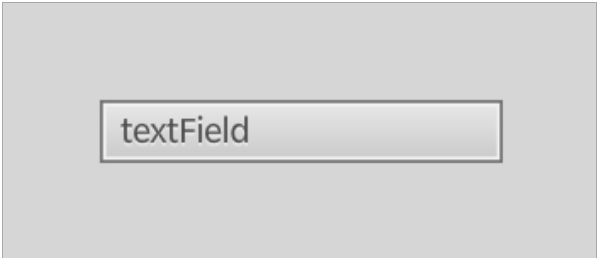


그림 6: 스킨 없는 버튼

CLIK Button
Button (gfx.controls.Button)
textField, (stats) Button
, ScrollBar Slider thumb
Button

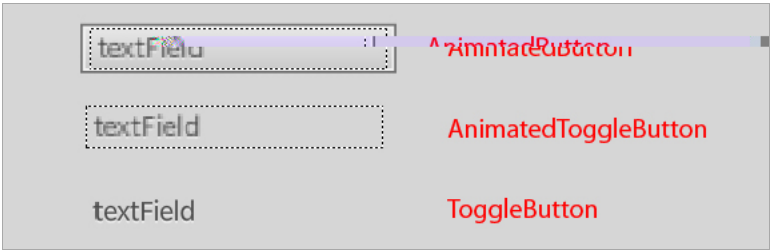
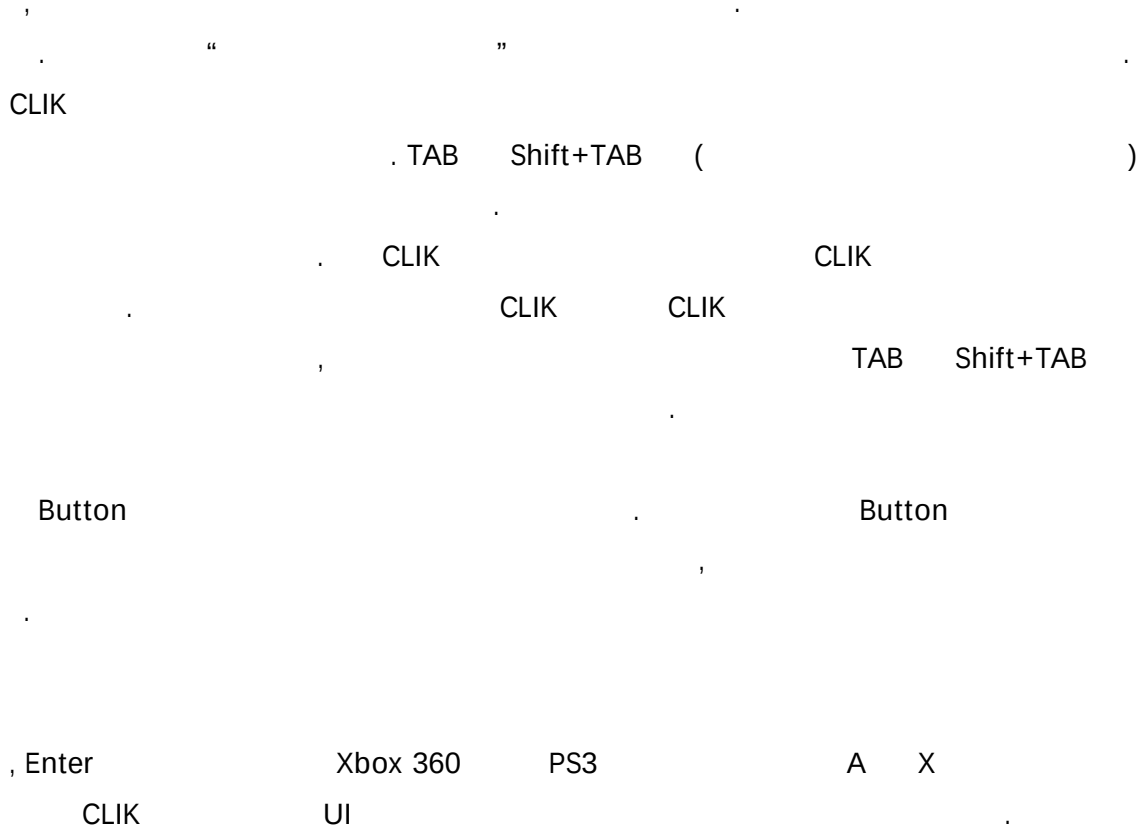


그림 7: AnimatedButton, AnimatedToggleButton, ToggleButton

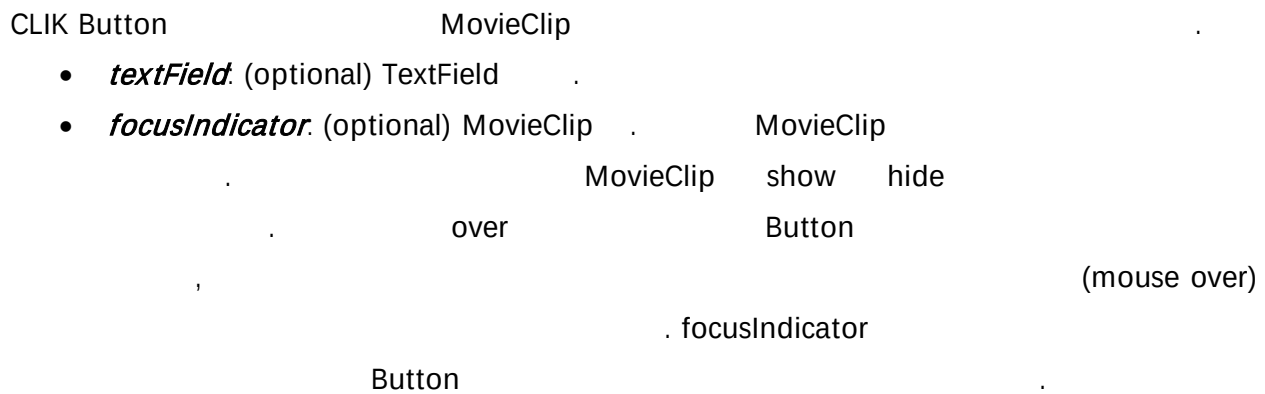
CLIK Button
Button.flas ToggleButton, AnimatedButton, AnimatedToggleButton

2.1.1.1 사용자 상호작용

Button



2.1.1.2 컴포넌트 설정



2.1.1.3 상태

CLIK

- *up*

- *over*
- *down*
- *disabled*

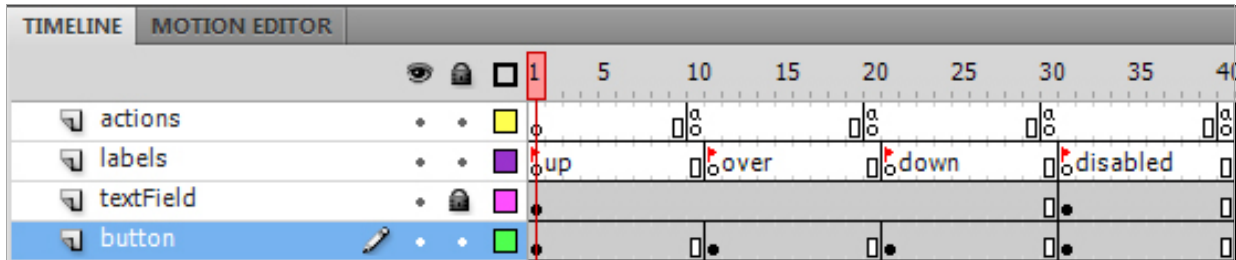


그림 8: Button 타임라인

Button

(transition)

[Getting Started with CLIK Buttons](#)

2.1.1.4 점검가능 속성

IDE

(component inspector panel)

. CS4

Window

Component inspector panel

[Shift+F7]

(inspectable properties)

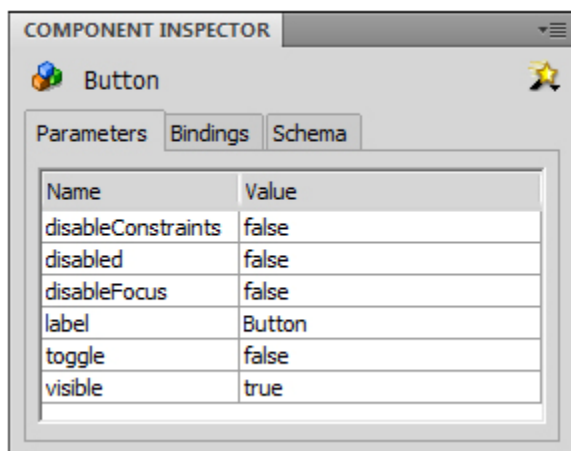


그림 9: CS4 component inspector 내의 Button 컴포넌트 점검가능 속성

Button

disableConstraints	Button	textField	true	
	disable	textField		,
		disable	textField	
	textField	/		
disabled	true	disable		diable
disableFocus				
	true			
label	Button			
toggle			. True	
visible	false			

IDE CLIK SWF

2.1.1.5 이벤트

CLIK

Object _____, _____.

- *type*.
- *target*.

Button

show	(visible property)	true	.
hide	(visible property)	false	.
focusIn	.		
focusOut	.		
select	(selected property)	.	
	<i>selected</i> .	, true.	(Boolean)
stateChange	.	.	
	<i>state</i> .	, “up”, “over”, “down”	.
			Getting Started with CLIK Buttons
rollOver	.	(rolled int)	
	<i>mouseIndex</i> .	(	
	).	. 0 3	
rollOut	.	(rolled out)	
	<i>mouseIndex</i> .	(	
	).	. 0 3	
press	.		
	<i>mouseIndex</i> .	(	
	).	. 0 3	
doubleClick	.	<i>doubleClickEnabled</i>	true
	.		
	<i>mouseIndex</i> .	(	
	).	. 0 3	
click	.		
	<i>mouseIndex</i> .	(	
	).	. 0 3	
dragOver	.	(	).
	<i>mouseIndex</i> .	(	
	).	. 0 3	

dragOut	(
<i>mouseIndex.</i>	(
).	. 0 3
releaseOutside	,
.	.
<i>mouseIndex.</i>	(
).	. 0 3

```
myButton.addEventListener("click", this, "onButtonPress");
function onButtonPress(event:Object) {
    // Do something
}
```

```
click, myButton,
onButtonPress
. ( event ) Object
Object,
```

2.1.1.6 팁과 트릭

```
Button :
import gdx.controls.Button;
attachMovie("Button", "btnInstanceName", someDepth, {_x:33,
_y:262, ...});
```

CLIK Button unselected selected

```
toggle :
btn.toggle = true;
```

```
:
btn.doubleClickEnabled = true;
btn.addEventListener("doubleClick", this, "onDoubleClick");
function onDoubleClick(e:Object):Void {
    // Button was double clicked!
}
```

click :

```
btn.autoRepeat = true;
```

2.1.2 CheckBox



그림 10: 스킨없는 CheckBox

CheckBox (gfx.controls.CheckBox) Button
· CheckBox true/false() · ToggleButton

2.1.2.1 사용자 상호작용

CheckBox
Button

2.1.2.2 컴포넌트 설정

CLIK CheckBox MovieClip

- **textField**: (optional) TextField
- **focusIndicator**: (optional) MovieClip · MovieClip
MovieClip show hide

2.1.2.3 상태

CheckBox selected

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

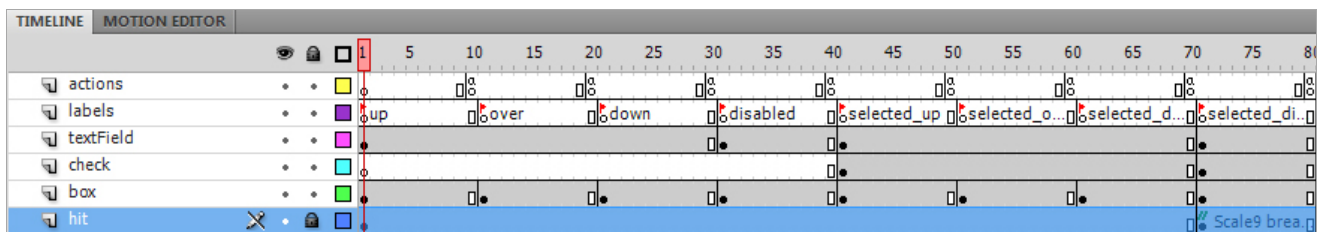


그림 11: CheckBox 타임라인.

CheckBox . Button ([Getting Started with CLIK Buttons](#))

2.1.2.4 점검가능 속성

Button CheckBox Button
, toggle disableFocus

Data	
disableConstraints	Button
	textField
	true
disable	
	textField
	. disable
	textField

	textField	/	.
disabled	true		
Label			
selected	true	CheckBox	()
Visible	false		

2.1.2.5 이벤트

Object , .

- *type*.
- *target*.

CheckBox .

Show	(visible property)	true	.
Hide	(visible property)	false	.
focusIn	.		
focusOut	.		
Select	(selected property)	.	
	<i>selected</i> .	(Boolean)	
stateChange	.		
	<i>state</i> .	.	“up”, “over”, “down”
	.		Getting Started with CLIK Buttons
rollOver	.		
	<i>mouseIndex</i> .	(	
	).	. 0 3	
Rollout	.		
	<i>mouseIndex</i> .	(	
	).	. 0 3	
Press	.		
	<i>mouseIndex</i> .	(	

	).	. 0	3	
doubleClick	.	<i>doubleClickEnabled</i>	true	
	.	<i>mouseIndex</i> .	(	
	).	. 0	3	
Click	.	<i>mouseIndex</i> .	(	
	).	. 0	3	
dragOver			(	).
	.	<i>mouseIndex</i> .	(	
	).	. 0	3	
dragOut			(	).
	.	<i>mouseIndex</i> .	(	
	).	. 0	3	
releaseOutside			,	
	.	. 0	3	
	.	<i>mouseIndex</i> .	(	
	).	. 0	3	

```

        CheckBox
        myCheckBox.addListener("select", this, "onCheckBoxToggle");
        function onCheckBoxToggle(event:Object) {
            // Do something
        }

```

2.1.3 Label



그림 12: 스킨없는 Label

CLIK Label (gfx.controls.Label) MovieClip
 (wrapping) textField .
 Label textField . Label textField , Label
 textField .
 Label MovieClip textField . , Label
 textField
 AS2 textField
 disabled .
 Label (constraints) Label
 (resizing) textField Label textField .
 textField Label
 textField .

2.1.3.1 사용자 상호작용

Label .

2.1.3.2 컴포넌트 설정

CLIK Label MovieClip .

- *textField*: TextField .

2.1.3.3 상태

CLIK Label disabled 2 .

- *default*
- *disabled*

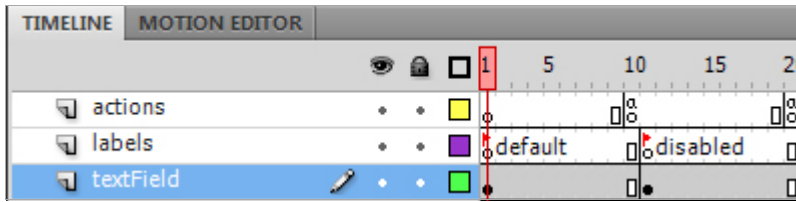


그림 13: Label 타임라인.

2.1.3.4 점검가능 속성

Label

text	
visible	false
disabled	true

2.1.3.5 이벤트

Object

- *type*.
- *target*.

Label

show	(visible property)	true
hide	(visible property)	false
stateChange	Label state: Label	“default” “disabled”

Label

```
myLabel.addEventListener("stateChange", this, "onStateChange");
function onStateChange(event:Object) {
    if (event.state == "default") {
        // Do something
    }
}
```

2.1.3.6 팁과 트릭

textField

HTML

8

lbl.htmlText = "foobar";

2.1.4 TextInput

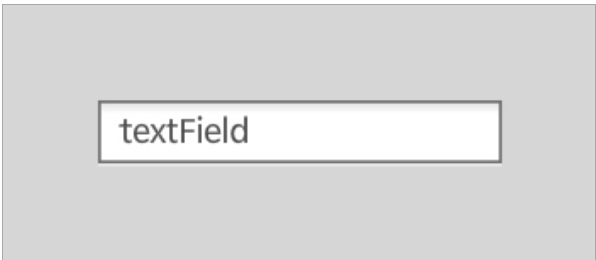
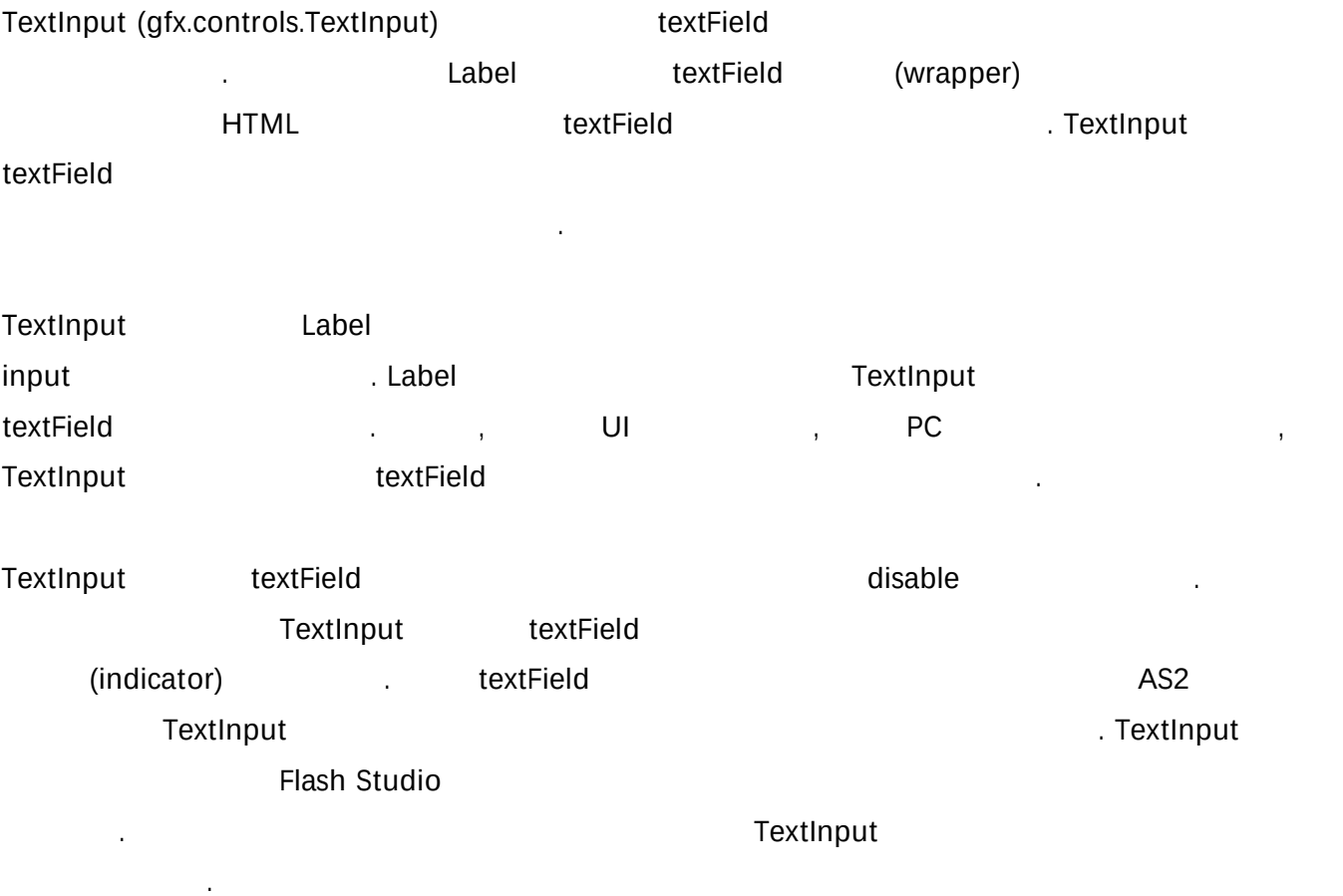


그림 14: 스킨없는 TextInput



TextInput	HTML	(multi paragraph)	textField
(Shift+ ,),	(Shift+Delete),	(Shift+Copy)	(Shift+Insert)

TextInput	roll over	roll out	
actAsButton			
			“over” “out”
rollOver rollOut		actAsButton	“over”
“out”	TextInput		“default”
	CLIK	TextInput	

	default Text
. defaultText	. Default Text (
) (0xAAAAAA)	
TextFormat defaultTextFormat	

2.1.4.1 사용자 상호작용

TextInput	textField
	textField

2.1.4.2 컴포넌트 설정

CLIK TextInput	MovieClip
----------------	-----------

- ***textField***: textField

2.1.4.3 상태

CLIK TextInput disabled 3

- *default* enabled
- *focused* , textField
- *disabled*

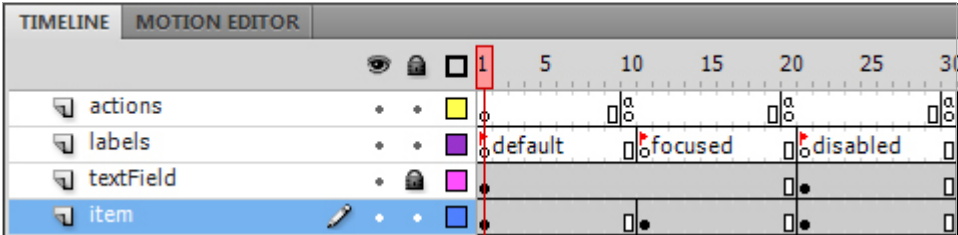


그림 15: TextInput 타임라인.

2.1.4.4 점검가능 속성

TextInput

text	textField
visible	false
disabled	true disable
editable	false TextInput
maxChars	textField (0)
password	true textField '*' . textField
defaultText	textField defaultTextFormat
actAsButton	true , TextInput rollover rollOut , TextInput

2.1.4.5 이벤트

Object

- *type*.
- *target*.

TextInput

show	(visible property)	true	.
hide	(visible property)	false	.
focusIn	.		
focusOut	.		
textChange	textField	.	
rollOver	actAsButton <i>mouseIndex</i> .	.	(
	)	0 3	.
rollOut	actAsButton <i>mouseIndex</i> .	.	(
	)	0 3	.

textField

```
myTextInput.addEventListener("textChange", this, "onTextChange");
function onTextChange(event:Object) {
    // Do something
}
```

2.1.4.6 팁과 트릭

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

TextInput HTML . textField HTML

8

```
textInput.htmlText = "<b>foo</b>bar";
```

2.1.5 TextArea

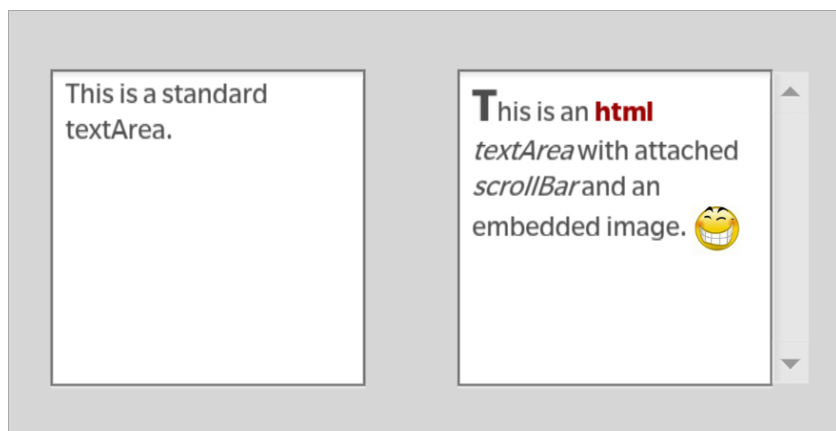


그림 16: 스킨없는 TextArea.

TextArea (gfx.controls.TextArea) CLIK TextInput

(ScrollBar) . TextInput TextArea
 TextInput .

Label TextInput TextArea multiline textField (wrapper) HTML
 (wrapping), , , textField
 . TextArea textField ,
 , TextArea .

2.1.5.1 사용자 상호작용

TextInput textField

2.1.5.2 컴포넌트 설정

CLIK TextArea

MovieClip

- *textField*: TextField type.

2.1.5.3 상태

TextInput

, TextArea

disabled

3

- *default* enabled
- *focused* , textField
- *disabled*

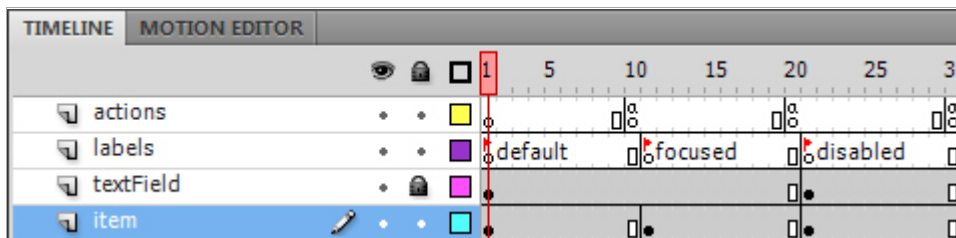


그림 17: TextArea 타임라인.

2.1.5.4 점검가능 속성

TextArea

TextInput

password

, CLIK ScrollBar

2.4

text	textField
visible	false
disabled	true disable .
editable	false TextInput .
maxChars	textField (0)
scrollBar	CLIK ScrollBar ScrollBar
	ID(TextArea)
scrollPolicy	auto

	. on ScrollBar (ScrollBar).	, off
defaultText	textField defaultTextFormat	.
actAsButton	true , TextInput rollOver rollOut , TextArea	.

2.1.5.5 이벤트

Object , .

- *type*.
- *target*.

TextArea

show	(visible property)	true	.
hide	(visible property)	false	.
focusIn	.		
focusOut	.		
textChange	textField		
scroll	.		
rollOver	actAsButton mouseIndex: (.) 0 3	.	
rollOut	actAsButton mouseIndex:	.	(

.) 0 3 .

```
TextArea
myTextArea.addEventListener("scroll", this, "onTextScroll");
function onTextScroll(event:Object) {
    // Do something
}
```

2.1.5.6 팁과 트릭

```
_global.gfxExtensions = true;
textInput.textField.noAutoSelection = true;
```

```
TextArea          HTML          textField          HTML
8
textInput.htmlText = "<b>foo</b>bar";
```

2.2 (Group Types)

```
RadioButton, ButtonGroup, ButtonBar          . ButtonGroup
Button
.
, ButtonBar
Button
. RadioButton
ButtonGroup          Button
```

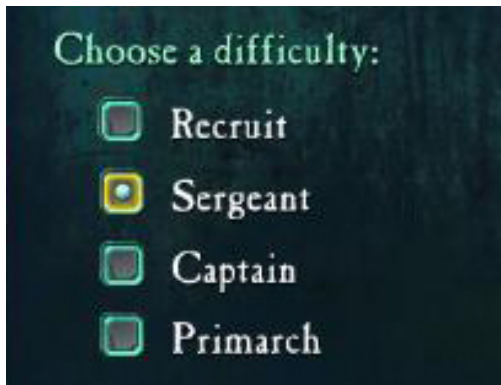


그림 18: *Dawn of War II*의 Radio ButtonGroup 예제

2.2.1 RadioButton

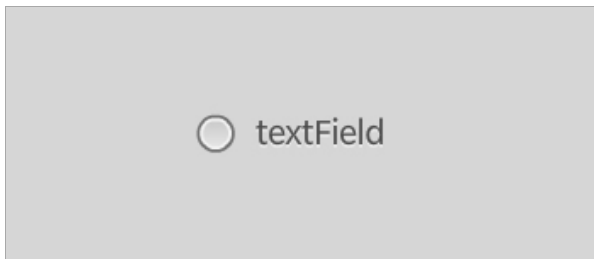


그림 19: 스킨없는 RadioButton

RadioButton (gfx.controls.RadioButton)

. RadioButton

RadioButton

CLIK RadioButton CheckBox

RadioButton

)

. RadioButton

ButtonGroup

,

ButtonGroup(

2.2.1.1 사용자 상호작용

```

        RadioButton
        .
        RadioButton
        ,
        RadioButton
        ButtonGroup
        ,
        RadioButton
        .
        RadioButton
        Button
        .

```

2.2.1.2 컴포넌트 설정

CLIK RadioButton MovieClip

- ***textField***: (optional) TextField .
- ***focusIndicator***: (optional) MovieClip . MovieClip
MovieClip show hide

2.2.1.3 상태

CheckBox RadioButton

- ***up***
- ***over*** ,
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over*** ,
- ***selected_down***
- ***selected_disabled***

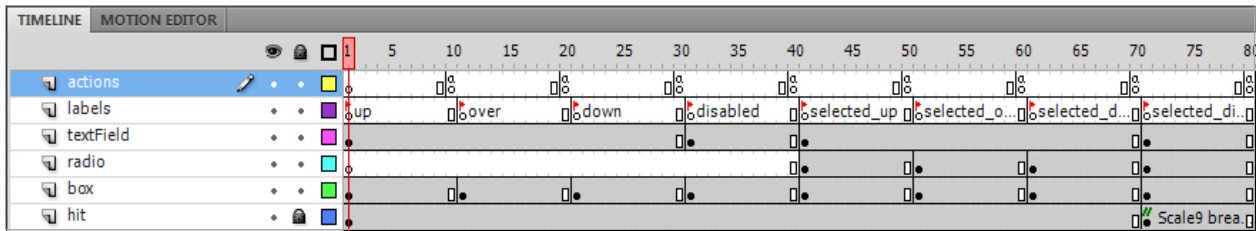


그림 20: RadioButton 타임라인.

RadioButton (RadioButton
[Getting Started with CLIK Buttons](#)

2.2.1.4 점검가능 속성

CheckBox Button toggle disableFocus

Label	
Visible	false
Disabled	true
disableConstraints	Button textField true disable textField , . disable textField , textField /
Selected	true RadioButton ()
Data	RadioButton label
Group	ButtonGroup RadioButton . RadioButton ButtonGroup RadioButton RadioButton _root ButtonGroup

<i>_root</i>	.	RadioButton
--------------	---	-------------

2.2.1.5 이벤트

Object , .

- *type*.
- *target*.

RadioButton .

Show	(visible property)	true	.
Hide	(visible property)	false	.
focusIn	.		
focusOut	.		
Select	(selected property)	.	
	<i>selected</i> .	(Boolean)	.
stateChange	.	.	.
	<i>state</i> .	.	“up”, “over”, “down”
	.		Getting Started with CLIK Buttons
rollOver	.	(rolled int)	.
	<i>mouseIndex</i> .	(	
	).	. 0 3	
Rollout	.	(rolled out)	.
	<i>mouseIndex</i> .	(	
	).	. 0 3	
Press	.	.	.
	<i>mouseIndex</i> .	(	
	).	. 0 3	
doubleClick	.	<i>doubleClickEnabled</i>	true

	<i>mouseIndex</i> .	(
	).	. 0 3
Click	.	
	<i>mouseIndex</i> .	(
	).	. 0 3
dragOver		().
	<i>mouseIndex</i> .	(
	).	. 0 3
dragOut		().
	<i>mouseIndex</i> .	(
	).	. 0 3
releaseOutside		,
	.	
	<i>mouseIndex</i> .	(
	).	. 0 3

```

RadioButton
myRadio.addListener("select", this, "onRadioToggle");
function onRadioToggle(event:Object) {
    // Do something
}

```

2.2.1.6 팁과 트릭

Group RadioButtons not on the same level:

Group RadioButton

```

import gfx.controls.ButtonGroup;
var myGroup:ButtonGroup = new ButtonGroup("groupName");
radio1.group = myGroup;
radio2.group = myGroup;
someMovie.radio1.group = myGroup;
someMovie.radio2.group = myGroup;

```

2.2.2 ButtonGroup

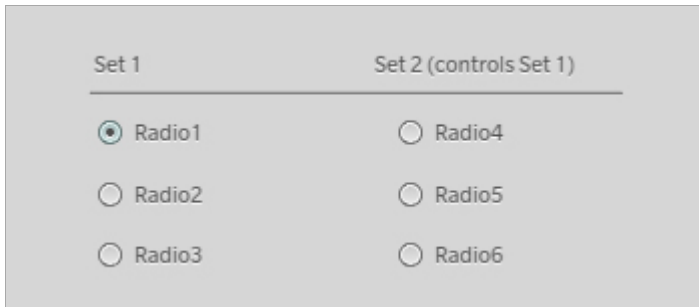


그림 21: 스킨없는 ButtonGroup.

CLIK ButtonGroup (gfx.controls.ButtonGroup)

Button (CheckBox RadioButton) . CLIK ButtonGroup

2.2.2.1 사용자 상호작용

ButtonGroup ButtonGroup
RadioButtons

2.2.2.2 컴포넌트 설정

CLIK ButtonGroup MovieClip
ButtonGroup

2.2.2.3 상태

ButtonGroup

2.2.2.4 점검가능 속성

ButtonGroup

2.2.2.5 이벤트

Object

- *type*.
- *target*.

ButtonGroup

change	• <i>item</i>. . CLIK Button • <i>data</i>. . AS2
itemClick	• <i>item</i>. . CLIK Button

ButtonGroup

```
myGroup.addEventListener("change", this, "onNewSelection");
function onNewSelection(event:Object) {
    if (event.item == myRadio) {
        // Do something
    }
}
```

2.2.2.6 팁과 트릭

RadioButton

```
var selectedRadio:Button = group.selectedButton;
```

RadioButtons

ButtonGroup

(listener)

```
radioBtn1.group.addEventListener("itemClick", this, "onClick");
```


2.2.3 ButtonBar

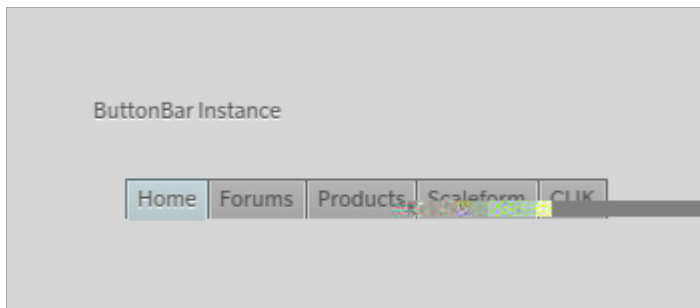


그림 22: 스킨없는 ButtonBar.

```
ButtonBar (gfx.controls.ButtonBar) ButtonGroup ,
    , dataProvider( ) Button
    . ButtonBar (tab-bar) UI
dataProvider
```

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4",
"item5"];
```

2.2.3.1 사용자 상호작용

```
ButtonBar ButtonGroup ,
    Button
```

2.2.3.2 컴포넌트 설정

```
CLIK ButtonBar MovieClip , ButtonBar
    MovieClip
```

2.2.3.3 상태

```
CLIK ButtonBar
    .
```

2.2.3.4 점검가능 속성

ButtonBar (IDE ButtonBar Button
) , .

visible	false	ButtonBar
disabled	true	ButtonBar
itemRenderer	Button	ID. ButtonBar
direction	. Horizontal Button	vertical
spacing	Button	direction (direction).
autoSize	true	Button
buttonWidth	Button	. autoSize

2.2.3.5 이벤트

Object , .

- *type*.
- *target*.

ButtonBar .

show	(visible property)	true	.
hide	(visible property)	false	.
focusIn			
focusOut			.

change

- *index*: ButtonBar . . 0
1
- *renderer*: . CLIK Button
- *item*: dataProvider . AS2
- *data*: dataProvider . AS2

itemClick

- *index*: Button ButtonBar . . 0
1
- *item*: dataProvider . AS2
- *data*: dataProvider . AS2
- *mouseIndex*: (. 0 3

ButtonBar Button

```
myBar.addEventListener("itemClick", this, "onItemClick");  
function onItemClick(event:Object) {  
    processData(event.data);  
    // Do something  
}
```

2.3 (Scroll Types)

ScrollIndicator, ScrollBar, Slider
 (thumb)

ScrollBar . ScrollBar 4

, (thumb grip), (track),

Slider ScrollBar thumb track

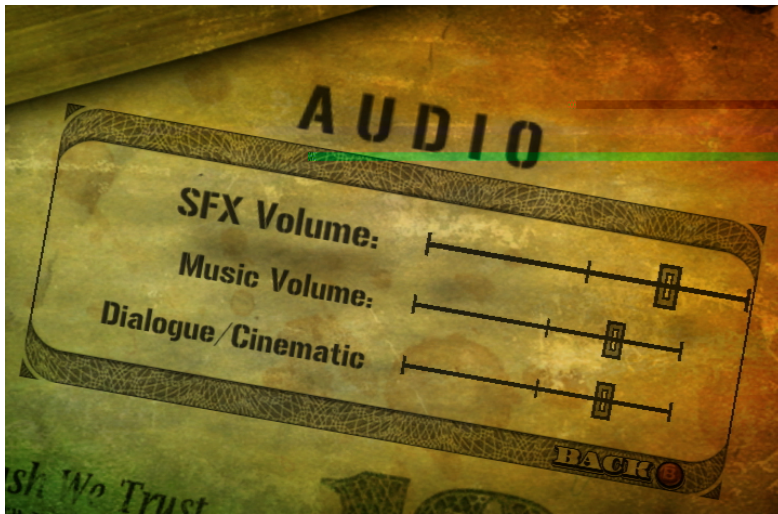


그림 23: *Mercenaries 2*의 자동 메뉴 Slider 예제

2.3.1 ScrollIndicator



그림 24: 스킨없는 ScrollIndicator.

CLIK ScrollIndicator(gfx.controls.ScrollIndicator) multiline textField
 textField

TextArea scrollbar ScrollIndicator

ScrollBar (ID)

2.3.1.1 사용자 상호작용

ScrollIndicator

2.3.1.2 컴포넌트 설정

CLIK ScrollIndicator

MovieClip

- *thumb*: CLIK
- *track*: MovieClip

2.3.1.3 상태

ScrollIndicator

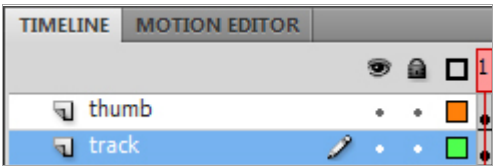


그림 25: ScrollIndicator 타임라인.

2.3.1.4 점검가능 속성

ScrollIndicator

scrollTarget	TextArea	textField	textField
			ScrollIndicator
visible	false		
disabled	true	disable	
offsetTop	(Thumb)		
offsetBottom			

2.3.1.5 이벤트

Object , .

```
scrollInd.direction = "horizontal";
```

2.3.2 ScrollBar

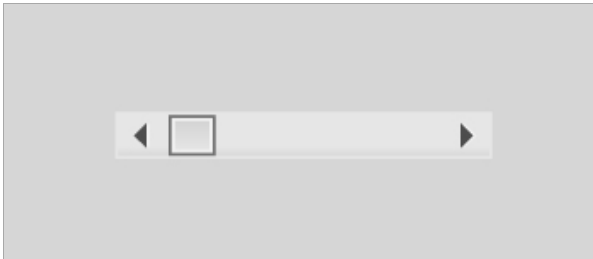


그림 26: 스킨없는 ScrollBar.

CLIK ScrollBar (gfx.controls.ScrollBar)

ScrollIndicator

2.3.2.1 사용자 상호작용

ScrollBar

ScrollBar
2
[SHIFT]

trackMode

. trackMode

2.3.2.2 컴포넌트 설정

CLIK ScrollBar

MovieClip

- **up** : CLIK Button . Button;
- **downArrow**: CLIK Button t . Button;
- **thumb**: CLIK Button .

- **track**: CLIK Button (scrollbar track) (girp)

2.3.2.3 상태

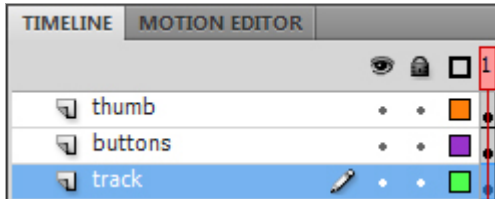


그림 27: ScrollBar timeline./ ScrollBar 타임라인

ScrollBar (thumb), up, down,

2.3.2.4 점검가능 속성

ScrollBar ScrollIndicator

scrollTarget	TextArea	textField	textField
	ScrollIndicator		
trackMode	scrollPage		
	scrollToCursor		
visible	false		
disabled	true	disable	
offsetTop	(Thumb)		
offsetBottom			

2.3.2.5 이벤트

Object

- *type*.
- *target*.

ScrollBar

Show	(visible property)	true	.
Hide	(visible property)	false	.
scroll	• <i>position</i>.		

```
mySB.addEventListener("scroll", this, "onListScroll");
function onListScroll(event:Object) {
    trace("scroll position: " + event.position);
    // Do something
}
```

2.3.2.6 팁과 트릭

```
scrollBar.setScrollProperties(pageSize, minPos, maxPos,
pageScrollSz);
scrollBar.positon = currPos;
```

```

        _rotation
        ScrollIndicator          horizontal
    ,
```

```
scrollInd.direction = "horizontal";
```

scrollPage

1

```
scrollBar.trackScrollPageSize = pageSize;
```

2.3.3 Slider

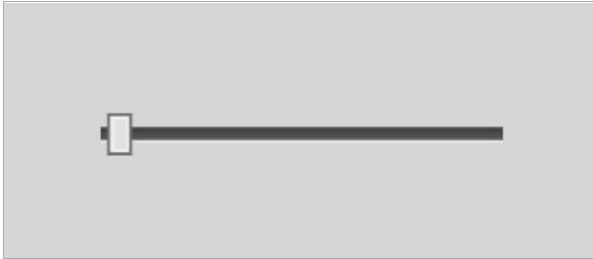


그림 28: 스킨없는 Slider.

Slider (gfx.controls.Slider)

2.3.3.1 사용자 상호작용

Slider

Slider

, [HOME], [END]

2

trackMode

trackMode

[SHIFT]

2.3.3.2 컴포넌트 설정

CLIK Slider

MovieClip

- **thumb**: CLIK Button
- **track**: CLIK Button (girp)

2.3.3.3 상태

ScrollBar ScrollIndicator Slider (thumb, track)

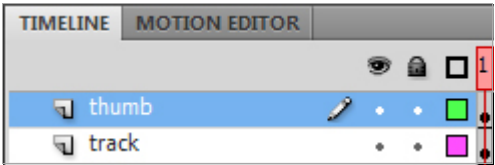


그림 29: Slider

2.3.3.4 점검가능 속성

Slider

visible	false	
disabled	true	disable
value	Slider	
minimum	Slider	
maximum	Slider	
snapping	true	snapInterval
snapInterval	snapping	false
liveDragging	true	Slider
	false	
offsetLeft		
offsetRight		

2.3.3.5 이벤트

Object ,

- *type*.
- *target*.

Slider

Show	(visible property)	true	.
Hide	(visible property)	false	.
focusIn	.		
focusOut	.		
change	.		

Slider

```
mySlider.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("slider value: " + event.target.value);
    // Do something
}
```

2.4 (List Types)

```
CLIK      NumericStepper, OptionStepper, ScrollingList, TileList,
DropdownMenu      . NumericStepper
      ,      DataProvider      . ListItemRenderer
      ,      ScrollingList      TileList
```

```

NumericStepper      OptionStepper      .      ScrollingList
TileList            .      2            ScrollIndicator
ScrollBar            .      DropdownMenu      (idle state)
                    ScrollingList      TileList

```


2.4.1 NumericStepper



그림 33: 스킨없는 Numeric Stepper.

NumericStepper (gfx.controls.NumericStepper)

2.4.1.1 사용자 상호작용

NumericStepper 2

[HOME] [END] step

2.4.1.2 컴포넌트 설정

NumericStepper MovieClip

- *textField*: TextField
- *nextBtn*: CLIK Button step
- *prevBtn*: CLIK Button step

2.4.1.3 상태

NumericStepper disabled 3

- *default* enabled
- *focused* , textField

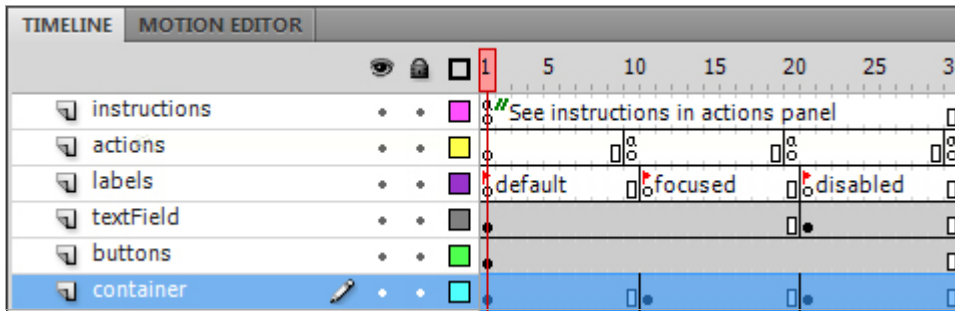


그림 34: NumericStepper 타임라인

2.4.1.4 점검가능 속성

NumericStepper MovieClip

visible	false	
disabled	true	disable
value	NumericStepper	
minimum	NumericStepper	
maximum	NumericStepper	

2.4.1.5 이벤트

Object

- *type*.
- *target*.

NumericStepper

Show	(visible property)	true
Hide	(visible property)	false
change	NumericStepper	
stateChange	NumericStepper	disabled
	• <i>state</i> .	“default”, “focused”, “disabled”

NumericStepper

```
myNS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("ns value: " + event.target.value);
    // Do something
}
```

2.4.1.6 팁과 트릭

```
ns.stepSize = 0.5;
```

```
ns.labelFunction = function(value:Number) {
    switch(value) {
        case:
            return value + "st";
        default:
            return value + "th";
    }
}
```

2.4.2 OptionStepper


```
optionStepper.dataProvider = ["item1", "item2", "item3", "item4"];
```

2.4.2.1 사용자 상호작용

NumericStepper OptionStepper 2

 [HOME] [END]

 dataProvider

2.4.2.2 컴포넌트 설정

CLIK NumericStepper MovieClip

textField: TextField

- *nextBtn*: CLIK Button dataProvider
- *prevBtn*: CLIK Button dataProvider

2.4.2.3 상태

OptionStepper disabled 3

- *default* enabled
- *focused* , textField
- *disabled*

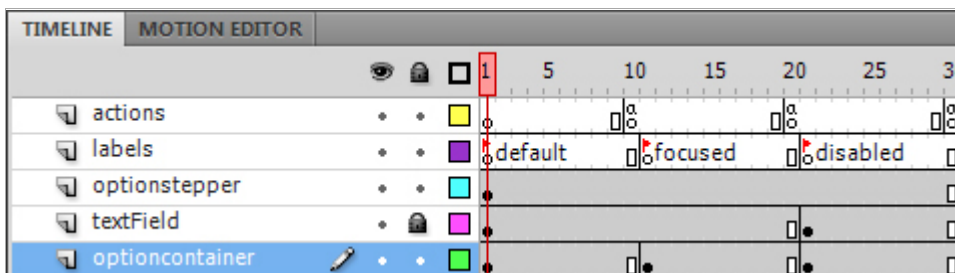


그림 36: OptionStepper 타임라인

2.4.2.4 점검가능 속성

OptionStepper

MovieClip

visible	false		
disabled	true	disable	.

2.4.2.5 이벤트

Object

- *type*.
- *target*.

OptionStepper

Show	(visible property)	true	.
Hide	(visible property)	false	.
change	OptionStepper	.	
stateChange	OptionStepper	disabled	.
	• <i>state</i>: "default", "focused", "disabled"	.	"default", "focused",

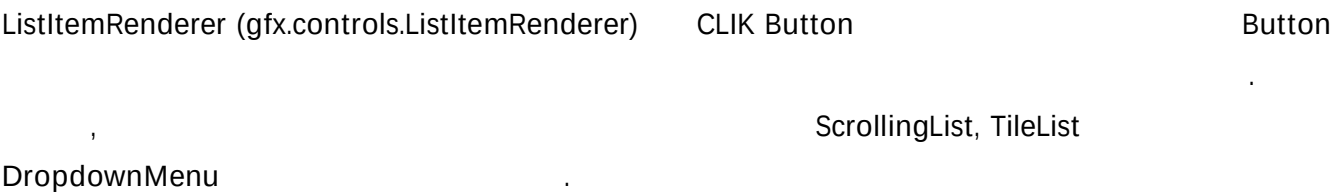
OptionStepper

```
myOS.addEventListener("change", this, "onValueChange");
function onValueChange(event:Object) {
    trace("os value: " + event.target.selectedItem);
    // Do something
}
```

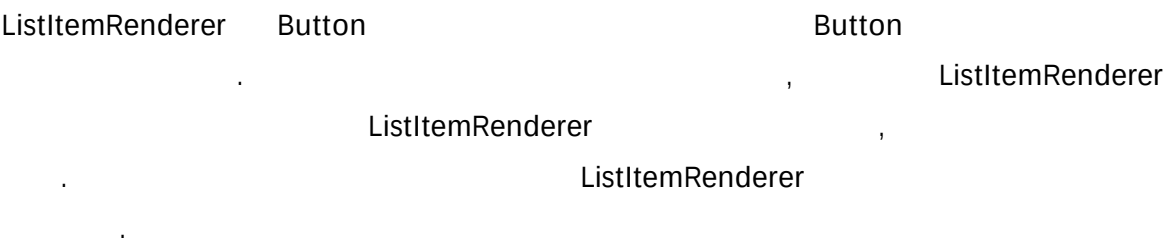
2.4.3 ListItemRenderer



그림 37: 스킨없는 ListItemRenderer.



2.4.3.1 사용자 상호작용



2.4.3.2 컴포넌트 설정



- **textField**. (optional) TextField
 - **focusIndicator**. (optional) MovieClip
- MovieClip show hide

2.4.3.3 상태

이 클래스는 `ListItemRenderer` 클래스의 `selected` 속성을 나타냅니다.

selected

- *up*
- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

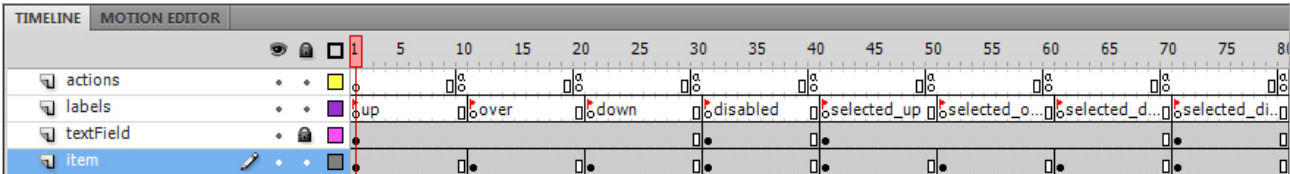


그림 38: `ListItemRenderer` 타임라인.

`ListItemRenderer` 클래스는 `Button` 클래스의 `selected` 속성을 나타냅니다. ([Getting Started with CLIK Buttons](#))

2.4.3.4 점검가능 속성

`ListItemRenderer` 클래스는 `Button` 클래스의 `selected` 속성을 나타냅니다.

Label	ListItemRenderer	.
visible	false	.
disabled	true	disable .

2.4.3.5 이벤트

Object , .

- *type*.
- *target*.

ListItemRenderer Button .

Show	(visible property)	true	.
Hide	(visible property)	false	
focusIn	.		
focusOut	.		
Select	(selected property)	.	
	• <i>selected</i> .	(Boolean)	
stateChange	.	.	"up", "over", "down"
	• <i>state</i> .		Getting Started with CLIK Buttons
Rollover	.		
	• <i>mouseIndex</i> .	(	
	).	. 0 3	
Rollout	.		
	• <i>mouseIndex</i> .	(	
	).	. 0 3	
Press	.		
	• <i>mouseIndex</i> .	(	
	).	. 0 3	
doubleClick	. <i>doubleClickEnabled</i>	true	
	.		
	• <i>mouseIndex</i> .	(	
	).	. 0 3	

Click	• <i>mouseIndex</i>. (0 3)
dragOver	• <i>mouseIndex</i>. (0 3)
dragOut	• <i>mouseIndex</i>. (0 3)
releaseOutside	• <i>mouseIndex</i>. (0 3)

2.4.4 ScrollingList



그림 39: 스킨없는 ScrollingList.

ScrollingList (gfx.controls.ScrollingList)

```
ScrollIndicator ScrollBar
dataProvider
```

```
scrollingList.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

ScrollingList ListItemRenderer . ListItemRenderer FLA
itemRenderer

2.4.4.1 사용자 상호작용

ScrollBar ScrollingList

ScrollingList

ScrollingList *wrapping* (

). *wrapping* normal

, *wrapping* wrap

wrap . *wrapping* stick

[Page Up] [Page Down]

[Home] [End] ,

ScrollBar

ScrollingList

ScrollBar

D-

CLIK

UI

2.4.4.2 컴포넌트 설정

ScrollingList

ScrollingList

2.4.4.3 상태

ScrollingList

disabled

3

- *default* enabled
- *focused* ,
- *disabled*

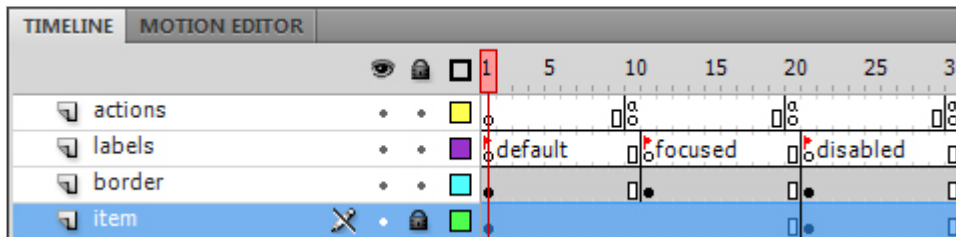


그림 40: ScrollingList 타임라인

2.4.4.4 점검가능 속성

ScrollingList	MovieClip	.
Visible	false	scrollbar
Disabled	false	disable disable ().
itemRenderer	ListItemRenderer	. <i>rendererInstanceName</i>
renderInstanceName	ScrollingList (prefix).	'r' 'r1', 'r2', 'r3',...
Scrollbar	ScrollBar	ScrollingList ScrollingList
Margin		<i>rendererInstanceName</i>

rowHeight	<i>rendererInstanceName</i>	.
paddingTop	<i>rendererInstanceName</i>	.
paddingBottom	<i>rendererInstanceName</i>	.
paddingLeft	<i>rendererInstanceName</i>	.
paddingRight	<i>rendererInstanceName</i>	.
thumbOffsetTop	.	.
thumbOffsetBottom	.	.
thumbSizeFactor	1.0	.
	1.0	.

2.4.4.5 이벤트

Object , .

- *type*.
- *target*.

ScrollingList .

Show	(visible property)	true	.
-------------	---------------------	------	---

Hide	(visible property)	false	.
focusIn	.		
focusOut	.		
Change	.		
	• <i>index</i>.	0	1
itemPress	.		
	• <i>renderer</i>.	. CLIK Button	
	• <i>item</i>.	.	
	dataProvider	. AS2	
	• <i>index</i>.	dataProvider	
	.	. 0	1
	• <i>mouseIndex</i>.	(	
	).	. 0	3
itemClick	..		
	• <i>renderer</i>.	. CLIK Button	.
	• <i>item</i>.	.	
	dataProvider	. AS2	.
	• <i>index</i>.	dataProvider	
	.	. 0	1
	• <i>mouseIndex</i>.	(	
	).	. 0	3
itemDoubleClick	.		
	• <i>renderer</i>.	. CLIK Button	
	• <i>item</i>.	.	
	dataProvider	. AS2	
	• <i>index</i>.	dataProvider	
	.	. 0	1
	• <i>mouseIndex</i>.	(	
	).	. 0	3
itemRollOver	.		
	• <i>renderer</i>.	. CLIK Button	
	• <i>item</i>.	.	
	dataProvider	. AS2	

- *index*. `dataProvider`
`. 0 1`
- *mouseIndex*. `(`
`). 0 3`

itemRollOut

- *renderer*. `. CLIK Button`
- *item*. `. AS2`
`dataProvider`
- *index*. `dataProvider`
`. 0 1`
- *mouseIndex*. `(`
`). 0 3`

```
myList.addEventListener("itemClick", this, "onItemClicked");
function onItemClicked(event:Object) {
    trace("list item was clicked: " + event.renderer);
    // Do something
}
```

2.4.4.6 팁과 트릭

```
// The ScrollingList automatically will use the property named
'label'
// if found in the item object, like so:
list.dataProvider = [{label: "one", data:1}, {label: "two",
data:2}];

// However if the item object has a different label property, such
as
// the following, then the list can be configured to use that
property
// instead:
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two",
data:2}];

// If the logic to construct a label from the item object is
```

```
// complicated and requires a function, then set the labelFunction
// property:
list.labelFunction = function(itemObj:Object):String {
    // Logic to construct a label
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];
```

2.4.5 TileList

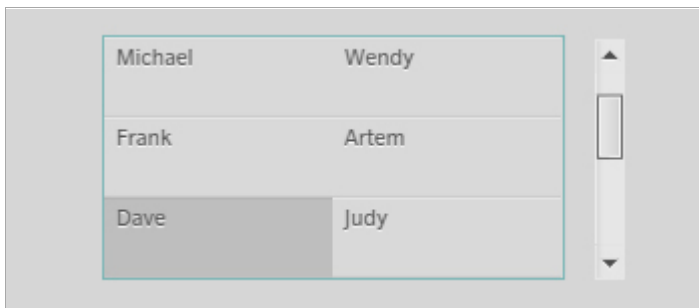


그림 41: 스킨없는 TileList.

TileList (gfx.controls.TileList) ScrollingList

```

        . ScrollIndicator    ScrollBar
        . TileList    ScrollingList
TileList    4
        .
        dataProvider    , dataProvider
```

```
tileList.dataProvider = ["item1", "item2", "item3", "item4",
"item5"];
```

```

TileList    ListItemRenderer    ListItemRenderer    FLA
        .
        itemRenderer
```

2.4.5.1 사용자 상호작용



[Page Up] [Page Down]

[Home] [End]



2.4.5.2 컴포넌트 설정



2.4.5.3 상태



- **default** enabled
- **focused** ,
- **disabled**

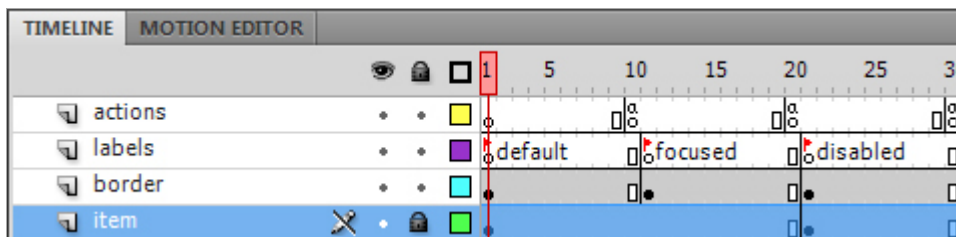


그림 42: TileList 타임라인

2.4.5.4 점검가능 속성

TileList

MovieClip

Visible	false	.	scrollbar
disabled	false	disable disable (	scrollbar
itemRenderer	ListItemRenderer	.	<i>rendererInstanceName</i>
renderInstanceName	ScrollingList (prefix).	.	'r' 'r1', 'r2', 'r3',... 1
Scrollbar	ScrollBar	.	ScrollingList ScrollingList
Margin	.	.	<i>rendererInstanceName</i>
rowHeight	.	.	<i>rendererInstanceName</i>
columnWidth	.	.	<i>rendererInstanceName</i>
externalColumnCount	.	.	<i>rendererInstanceName</i>

2.4.5.5 이벤트

Object , .

- *type*.
- *target*.

TileList .

Show	(visible property)	true	.
Hide	(visible property)	false	.
focusIn	.		
focusOut	.		
Change	.		
	• <i>index</i> .	.	0
	1		
itemPress	.		
	• <i>renderer</i> .	.	CLIK Button
	• <i>item</i> .	.	
	dataProvider	.	AS2
	• <i>index</i> .	dataProvider	
	.	0	1
	• <i>mouseIndex</i> .	(	
	).	0	3
itemClick	..		
	• <i>renderer</i> .	.	CLIK Button
	• <i>item</i> .	.	
	dataProvider	.	AS2
	• <i>index</i> .	dataProvider	
	.	0	1
	• <i>mouseIndex</i> .	(	
	).	0	3
itemDoubleClick	.		

- *renderer:* . CLIK Button
- *item:* .
dataProvider . AS2
- *index:* dataProvider
. 0 1
- *mouseIndex:* (. 0 3)

itemRollOver

- *renderer:* . CLIK Button
- *item:* .
dataProvider . AS2
- *index:* dataProvider
. 0 1
- *mouseIndex:* (. 0 3)

itemRollOut

- *renderer:* . CLIK Button
- *item:* .
dataProvider . AS2
- *index:* dataProvider
. 0 1
- *mouseIndex:* (. 0 3)

```

TileList
myList.addEventListener("focusIn", this, "onListFocused");
function onListFocused(event:Object) {
    trace("tile list was focused!");
    // Do something
}

```

2.4.5.6 팁과 트릭

```

// The TileList automatically will use the property named 'label'
// if found in the item object, like so:

```



```

list.dataProvider = [{label: "one", data:1}, {label: "two",
data:2}];

// However if the item object has a different label property, such
as
// the following, then the list can be configured to use that
property
// instead:
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two",
data:2}];

// If the logic to construct a label from the item object is
// complicated and requires a function, then set the labelFunction
// property:
list.labelFunction = function(itemObj:Object):String {
    // Logic to construct a label
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

2.4.6 DropdownMenu

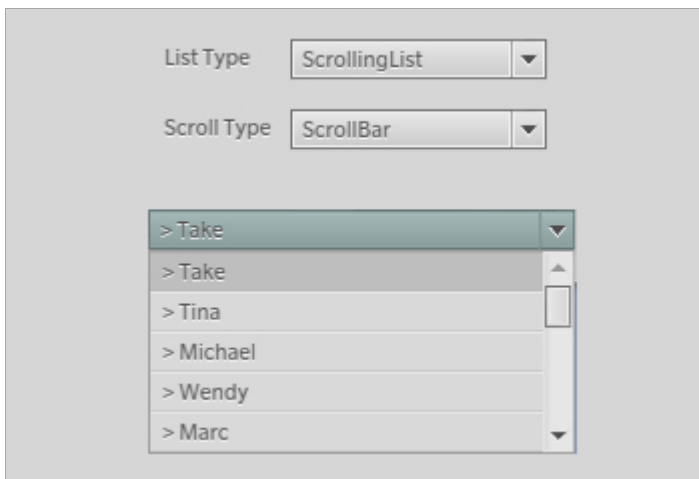


그림 43: 스킨없는 DropdownMenu.

DropdownMenu (gfx.controls.DropdownMenu)

```

. DropdownMenu (idle state)
. ScrollingList TileList , ScrollBar
ScrollIndicator . DropdownMenu dataProvider
, dataProvider

```

```

dropdownMenu.dataProvider = ["item1", "item2", "item3", "item4"];

```

DropdownMenu ScrollingList
 ListItemRenderer FLA dropdown

 , DropdownMenu scrollbar Scrollbar
 ScrollIndicator

2.4.6.1 사용자 상호작용

DropdownMenu [Enter] ScrollingList,
 TileList, ScrollBar

 , DropdownMenu .
 DropdownMenu

2.4.6.2 컴포넌트 설정

DropdownMenu Button
 DropdownMenu MovieClip

- ***textField***: (optional) TextField
- ***focusIndicator***: (optional) MovieClip MovieClip
 MovieClip show hide

2.4.6.3 상태

DropdownMenu ToggleButton CheckBox

- ***up***
- ***over***
- ***down***
- ***disabled***

- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

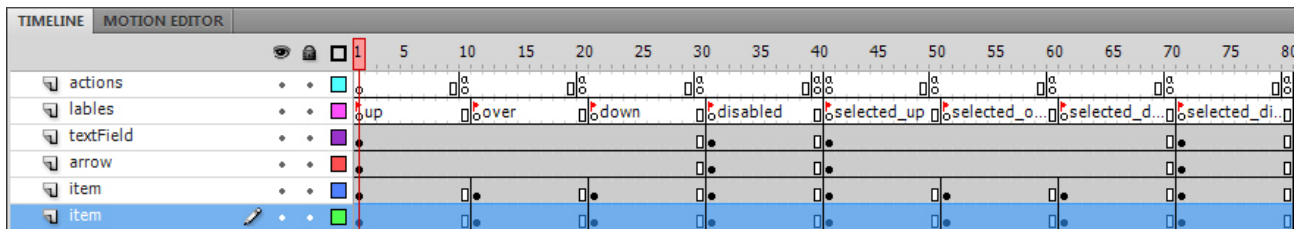


그림 44: DropdownMenu 타임라인

DropdownMenu . Button ([Getting Started with CLIK Buttons](#))

2.4.6.4 점검가능 속성

DropdownMenu

Visible	false
Disabled	true
Dropdown	DropdownMenu ((ScrollingList TileList)
dropdownWidth	-1 DropdownMenu
itemRenderer	
Scrollbar	
Margin	
paddingTop	
paddingBottom	

paddingLeft			
paddingRight			
thumbOffsetTop			
thumbOffsetBottom			
thumbSizeFactor1.0			
Offset	dropdown	dropdown	
	dropdown		
Offset	dropdown	dropdown	
Extent	offsetX		
	dropdownWidth	-1	
Direction		“up”	“down”

2.4.6.5 이벤트

Object			
<i>type</i>. <i>target</i>.			
DropdownMenu		change	Button

Show	(visible property)	true
Hide	(visible property)	false

focusIn	.
focusOut	.
Change	• <i>index</i>: . 0 1 • <i>data</i>: . AS2
Select	(selected property) .
	• <i>selected</i> : Button . (Boolean)
stateChange	• <i>state</i>: . , “up”, “over”, “down” Getting Started with CLIK Buttons
Rollover	• <i>mouseIndex</i>: (). . 0 3
Rollout	• <i>mouseIndex</i>: (). . 0 3
Press	• <i>mouseIndex</i>: (). . 0 3
doubleClick	<i>doubleClickEnabled</i> true • <i>mouseIndex</i>: (). . 0 3
Click	• <i>mouseIndex</i>: (). . 0 3
dragOver	() . • <i>mouseIndex</i>: (). . 0 3
dragOut	() . • <i>mouseIndex</i>: (

3.0).

releaseOutside

- *mouseIndex*: (

2.4.6.6 팁과 트릭

```
dropdown.addEventListener("click", this, "onClick");
function onClick(e:Object) {
    if (e.target.isOpen) {
        // Do something when open
    }
}
```

DropDownMenu

```
// Make sure that the linkage IDs used in the code are available;
// meaning that the symbols/components referred to by them exist
// in the .fla's library. For example: The following code requires
the
// DropdownMenu, ScrollingList and ScrollBar components.
attachMovie("DropdownMenu", "dd", this.getNextHighestDepth(),
            {dropdown: "ScrollingList"});
dd.dataProvider = ["one", "two", "three", "four", "five", "six"];

// Installing a scroll bar or scroll indicator with the dropdown's
// list is a little complicated, as it requires a delayed property
// set using trick presented in the following code. The delay is
required
// to allow the dropdown instantiate the list first, before
attaching
// a scroll bar to the list:
onEnterFrame = function() {
    dd.dropdown.scrollBar = "ScrollBar";
    onEnterFrame = null;
}
```

```
// The DropdownMenu automatically will use the property named
'label'
// if found in the item object:
```

```

list.dataProvider = [{label: "one", data:1}, {label: "two",
data:2}];

// However if the item object has a different label property, such
as
// the following, then the list can be configured to use that
property
// instead:
list.labelField = "name";
list.dataProvider = [{name: "one", data:1}, {name: "two",
data:2}];

// If the logic to construct a label from the item object is
// complicated and requires a function, then set the labelFunction
// property:
list.labelFunction = function(itemObj:Object):String {
    // Logic to construct a label
}
list.dataProvider = [{p1: "foo", p2: 1}, {p1: "bar", p2: 2}];

```

2.5 (Progress Types)

CLIK 2 : StatusIndicator Progress Bar.
 StatusIndicator , ProgressBar StatusIndicator

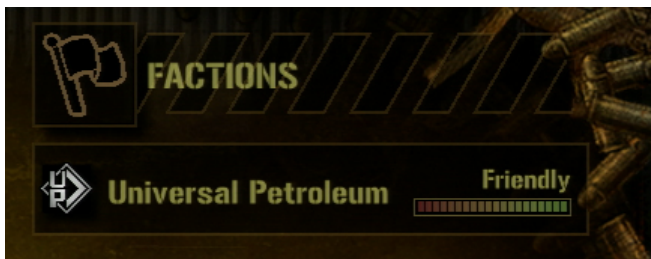


그림 45: *Mercenaries 2*의 팩션 StatusIndicator 사용예

2.5.1 StatusIndicator

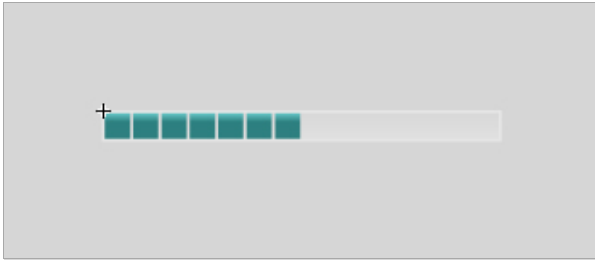


그림 46: 스킨없는 StatusIndicator.

```
StatusIndicator (gfx.controls.StatusIndicator)
    (indicator) . StatusIndicator
```

2.5.1.1 사용자 상호작용

```
StatusIndicator .
```

2.5.1.2 컴포넌트 설정

```
StatusIndicator MovieClip . , StatusIndicator
    2 . stop()
    . StatusIndicator value
gotoAndStop()
```

2.5.1.3 상태

```
StatusIndicator .
```

2.5.1.4 점검가능 속성

```
StatusIndicator MovieClip .
```

visible	false	.
disabled	true	disable .
value		.
minimum		.

maximum

2.5.1.5 이벤트

Object

- *type*.
- *target*.

StatusIndicator

show	(visible property)	true	.
hide	(visible property)	false	.

2.5.2 ProgressBar

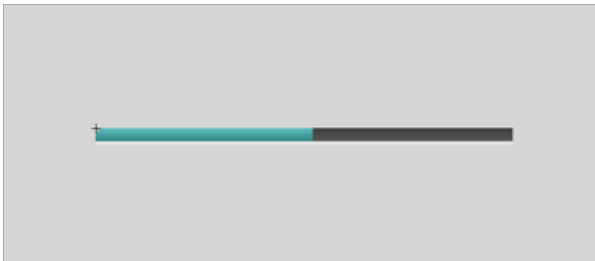


그림 47: 스킨없는 ProgressBar.

ProgressBar (gfx.controls.ProgressBar) StatusIndicator

ProgressBar ((bytesLoaded bytesTotal)

2.5.2.1 사용자 상호작용

ProgressBar

2.5.2.2 컴포넌트 설정

```
StatusIndicator , ProgressBar MovieClip
, ProgressBar 2 . stop()
. ProgressBar value
gotoAndStop()
```

2.5.2.3 상태

ProgressBar

2.5.2.4 점검가능 속성

ProgressBar MovieClip

visible	false				
disabled	true		disable		
target	ProgressBar	bytesLoaded	bytesTotal	" (listen)"	
mode	ProgressBar		"manual"		
	setProgress			"polled"	
	bytesLoaded	bytesTotal		"event"	
	bytesLoaded	bytesTotal		progress	

2.5.2.5 이벤트

Object

- *type*.
- *target*.

ProgressBar

show	(visible property)	true	.
hide	(visible property)	false	.
progress	ProgressBar		.
complete	ProgressBar		.

```
myProgress.addEventListener("progress", this, "onProgress");
myProgress.addEventListener("complete", this, "onProgress");
function onProgress(event:Object) {
    if (event.type == "progress") {
        // Do something
    } else
    {
        trace("Loading complete!");
    }
}
```

2.6 (Other Types)

CLIK
 UI
 Dialog, UILoader,
 ViewStack
 Dialog
 UILoader
 , ViewStack

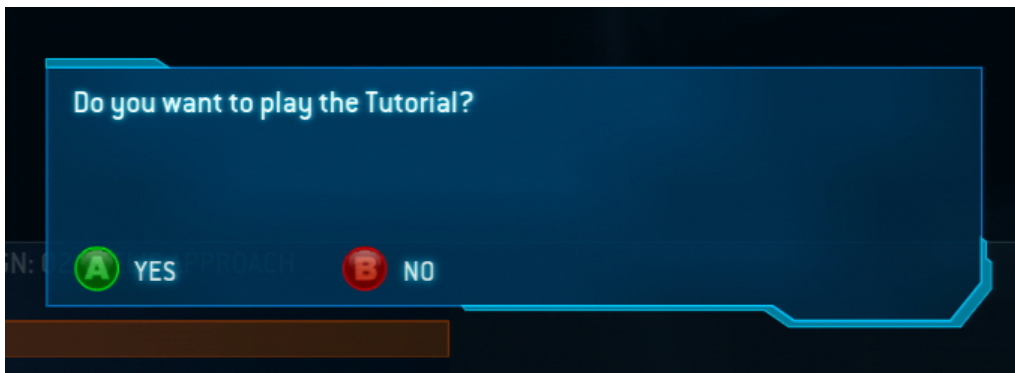


그림 48: *Halo Wars*의 Dialog 사용 예제

2.6.1 Dialog

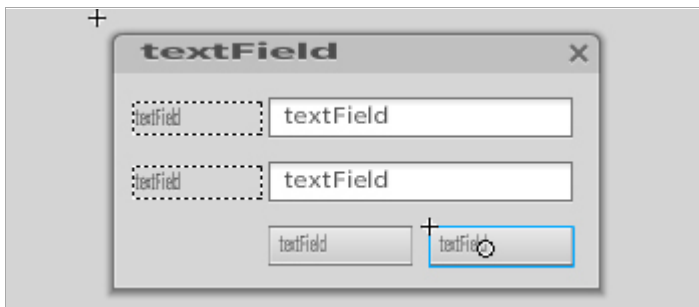


그림 49: 스킨없는 Dialog 샘플

Dialog (gfx.controls.Dialog)

MovieClip
 , MovieClip ()
 show hide

Dialog.show()

2.6.1.1 사용자 상호작용

Dialog

2.6.1.2 컴포넌트 설정

Dialog

MovieClip

- ***closeBtn***: (optional) CLIK Button . close
- ***cancelBtn***: (optional) CLIK Button . cancel
- ***submitBtn***: (optional) CLIK Button . OK
- ***dragBar***: (optional) MovieClip

2.6.1.3 상태

Dialog

MovieClip

2.6.1.4 점검가능 속성

Dialog

MovieClip

visible	false	
disabled	true	disable

2.6.1.5 이벤트

Object

- *type*.
- *target*.

Dialog

show		(visible property)	true	.
hide		(visible property)	false	.
submit	Dialog	(submit)	.	
	• <i>data</i> .	Dialog	getSubmitData	
		(Boolean)true	getSubmitData	AS2
close	Dialog	close	.	

(submit)

```
myDialog.addEventListener("submit", this, "onLoginSubmit");
function onLoginSubmit(event:Object) {
    trace("Received data from dialog: " + event.data);
}
```

2.6.2 UILoader

UILoader (gfx.controls.UILoader) SWF/GFX

UILoader

Scaleform

2.6.2.1 사용자 상호작용

UILoader SWF/GFX UILoader

2.6.2.2 컴포넌트 설정

UILoader

MovieClip

- **bg**: (optional) MovieClip . UILoader

2.6.2.3 상태

UILoader

. UILoader SWF/GFX

2.6.2.4 점검가능 속성

UILoader

MovieClip

autoSize	true	UILoader
maintainAspectRatio	true	UILoader
	false	UILoader
Source	SWF/GFX	
Timeout		UILoader ioError

2.6.2.5 이벤트

Object

- **type**:
- **target**:

UILoader

show	(visible property)	true
hide	(visible property)	false

progress	a)	b)	
loaded		0	100
complete			
ioError	source		

```
myUILoader.addEventListener("ioError", this, "onLoadingError");
function onLoadingError(event:Object) {
    displayErrorMessage("Could not load" + event.target.source);
}
```

2.6.3 ViewStack

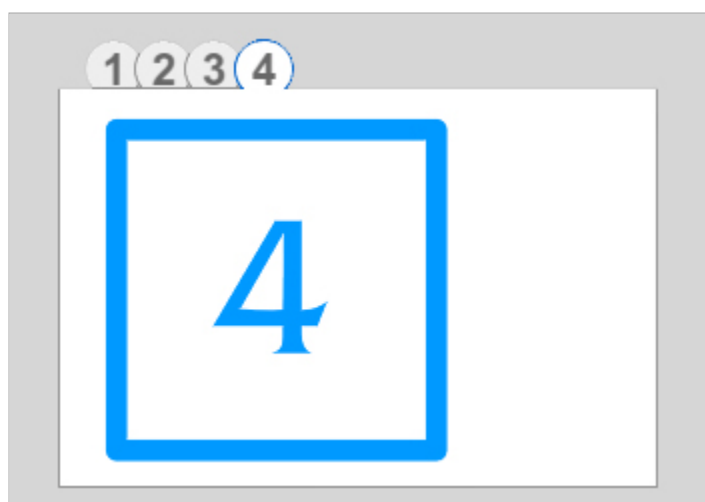


그림 50: 스킨없는 ViewStack.

ViewStack (gfx.controls.ViewStack)

```
TabBox    Accordion(
    . ViewStack    RadioButton
```


2.6.3.1 사용자 상호작용

ViewStack . ViewStack
View .

2.6.3.2 컴포넌트 설정

ViewStack MovieClip . ,

2.6.3.3 상태

ViewStack . ViewStack View

2.6.3.4 점검가능 속성

ViewStack MovieClip .

visible	false	
cache	true	. ViewStack
	targetGroup	.
targetGroup	'change'	(ButtonGroup) .
		ID data . RadioButton
		, IDE Component Inspector ID
	data	.

2.6.3.5 이벤트

ViewStack .

3 아트 세부설명

CLIK

3.1

CLIK

3.1.1 픽셀퍼펙트 이미지(Pixel-perfect Images)

CLIK

1. View
2. Grid Show Grid
3. View
4. Grid Edit Grid
5. , 1px
6. OK

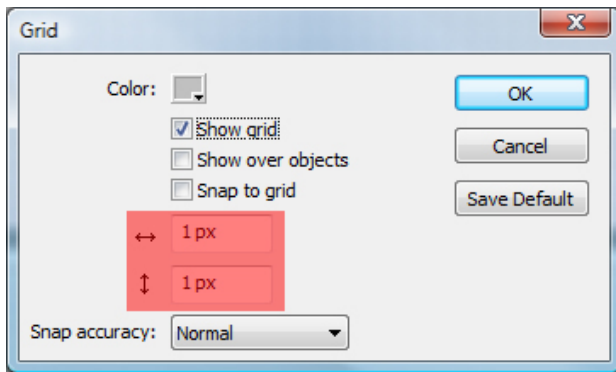


그림 51: Edit Grid 창.

1
SWF (blur)

2
“2”

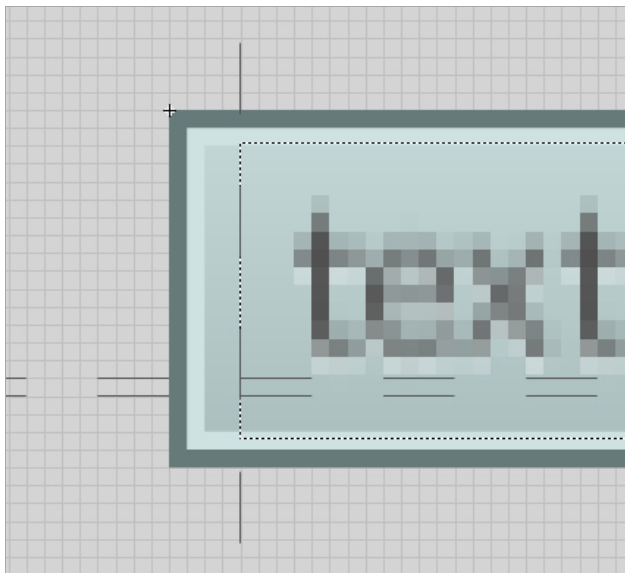


그림 52: 픽셀퍼펙트 벡터 그래픽 스킨

3.1.2 마스크(Masks)

(single digit)

PNG

3.1.3 애니메이션(Animations)

tessellation()
tessellation

tween tween tween
tween tween

3.1.4 레이어와 그리기 기본요소(Layers and Draw Primitives)

3.1.5 복잡한 스킨

(blurring)

3.1.6 2의 제공

- 16x16

- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

CLIK

3.2.1 컴포넌트 복제

2

3.2.1.1 컴포넌트 복제 (방법 1)

() CLIK (Button) FLA
FLA

1. *CLIK_Components.fla*

2. (Button) FLA
[*Copy*], FLA [*Paste*]

3. FLA *Rename* 'Button'

4. FLA *Properties*

5. *Identifier* 3

6. *Paste*

3.2.1.2 컴포넌트 복제 (방법 2)

1.  
2. *Properties* (: gfx.controlsButton) *Class*
[CTRL+C]
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export for ActionScript*
6. *Class* [CTRL+V]
7. *OK*
8. *Component Definition*
9. *Class* [CTRL+V]    

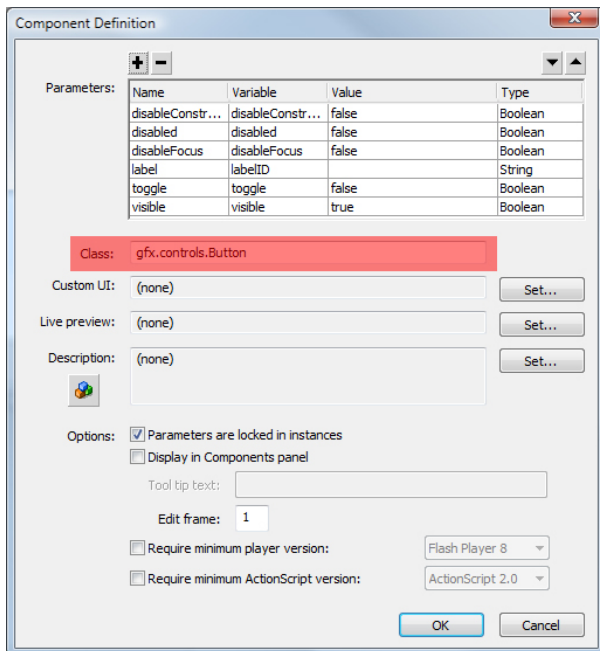


그림 54: 컴포넌트 정의 (클래스 부분을 채워 넣어야 한다).

3.3

CLIK

(look and feel)

CLIK

CLIK

. FLA

-
-
-

[Getting Started with CLIK](#)

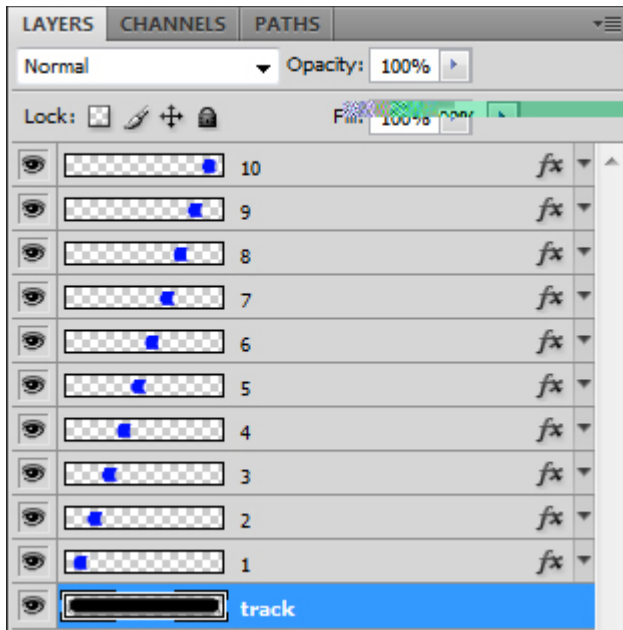


그림 57: StatusIndicator PSD 파일의 포토샵 레이어 설정

1. StatusIndicator
2. PSD
3. File Import Import to Stage
4. StatusIndicator PSD
5. Import Convert layers to Layers
6. OK
7. PSD

10	,	PSD	10
(		1	10
1	,	10	1
- 8.
- 9.
10. 1 6
11. 1 11 Insert Keyframe
12. 2 [CTRL+X]
13. 1 11 [CTRL+SHIFT+V]
14. 10 (1) 5

2 11, 3 16,
 4 21 10 PSD ,
 51
 15. 2~10
 16. , 5
Remove Frames
 55
 17. *indicator*
 18. *track track*
 19. 1 indicator
 20. indicator track actions , track
 indicator

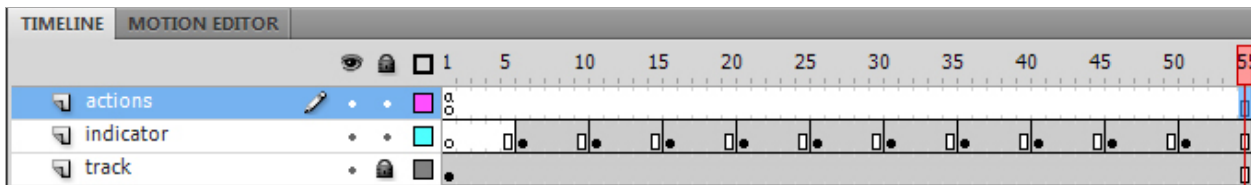


그림 58: The 최종 타임라인 (키프레임 위치와 최종 프레임의 위치를 확인하자).

21. StatusIndicator
 22. Inspectable parameter 1 10
 23.
 24.

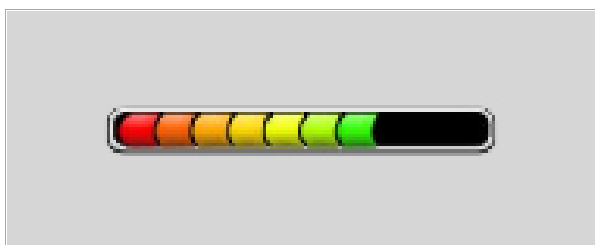


그림 59: 스킨있는 StatusIndicator.

3.4 Localization

CLIK

3.4.1 개요

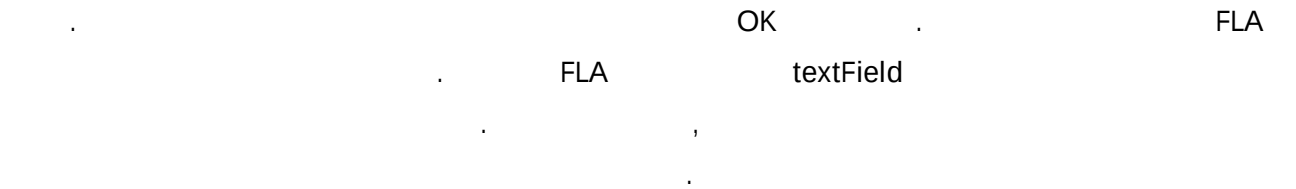
Scaleform () SWF
ScaleformLocalization System . UI

3.4.2 폰트 내장

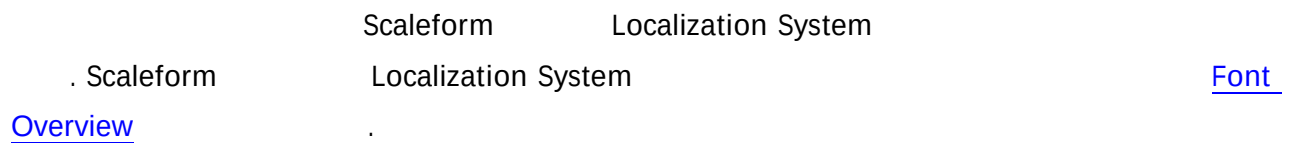
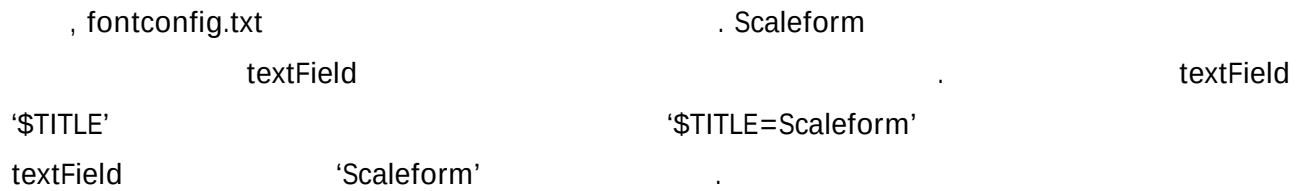
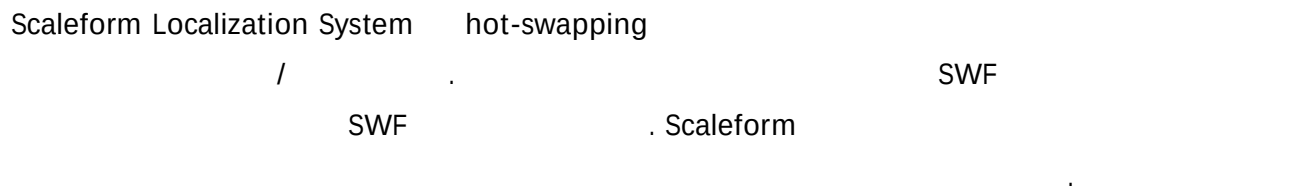
Scaleform
. Scaleform
Scaleform
Slate Mobile textField
(CJK)
ASCII Slate Mobile CJK
textField Scaleform Localization System (3.4.4)
Scaleform Localization System
Localization
. UI localization

3.4.3 textField 에 폰트 내장

textField
()
textField
textField (Window Property Inspector) *Embed*



3.4.4 Scaleform Localization System



4 프로그래밍 세부사항

, CLIK

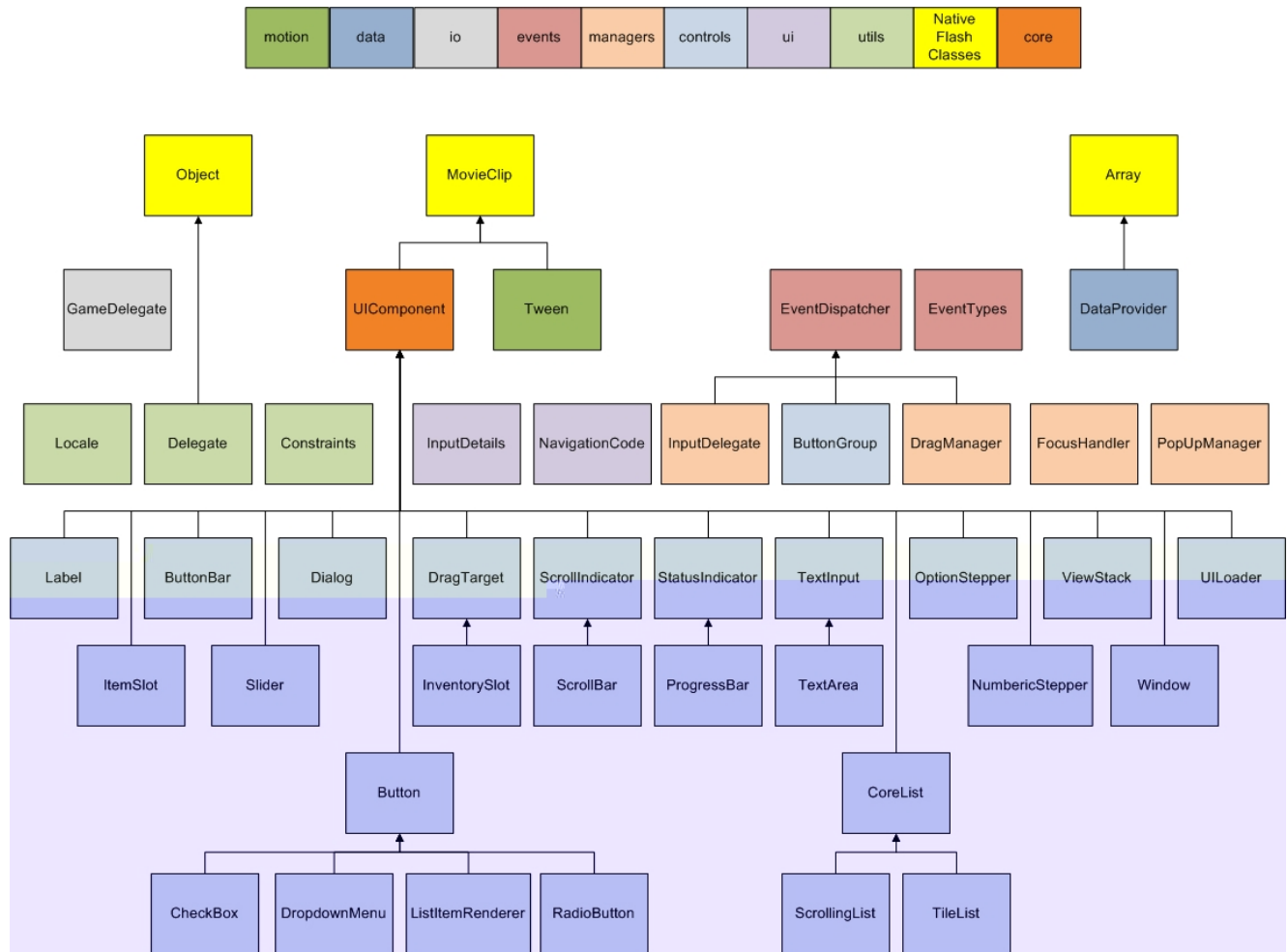
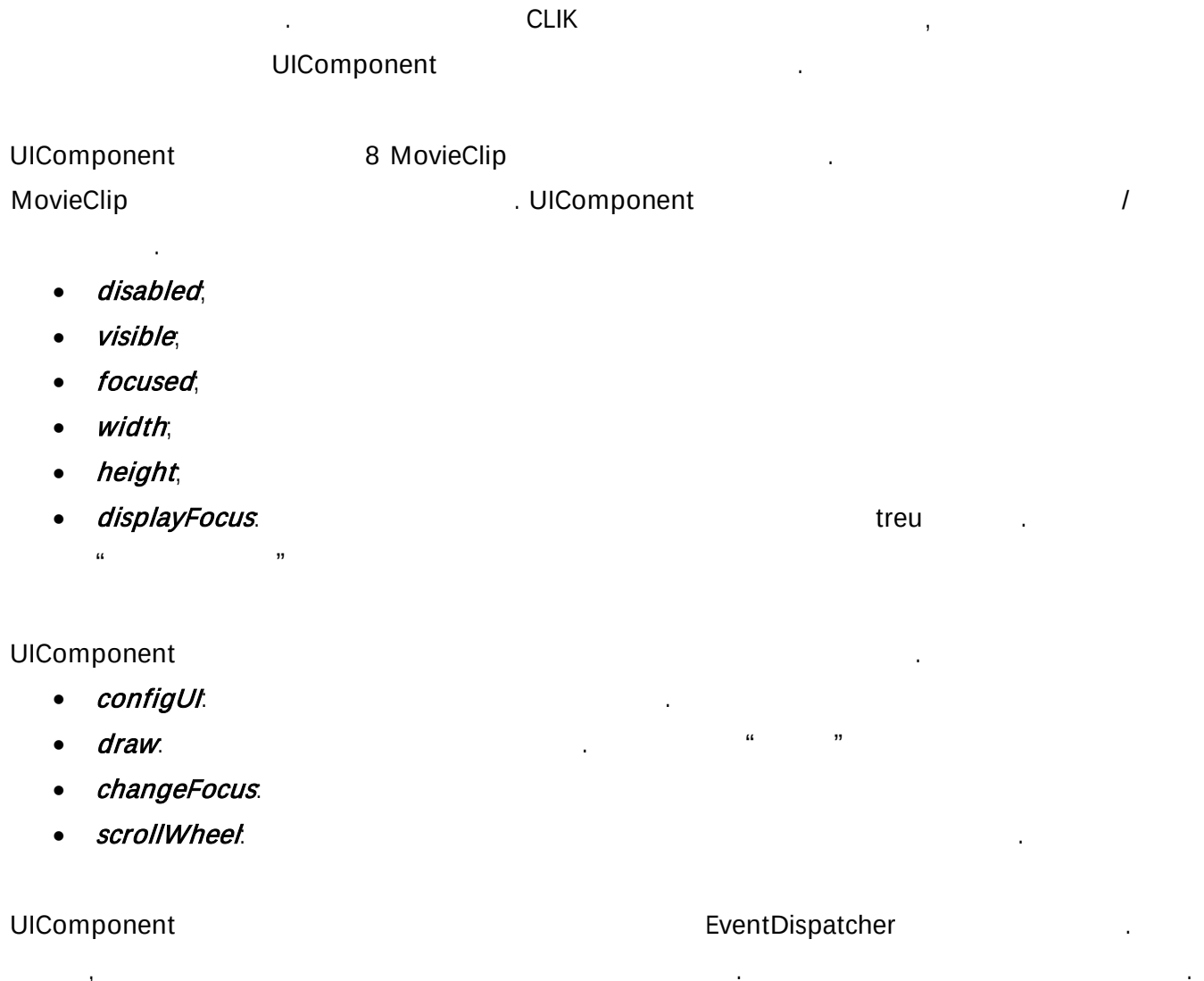


그림 60: 스케일폼 CLIK 클래스 계층구조

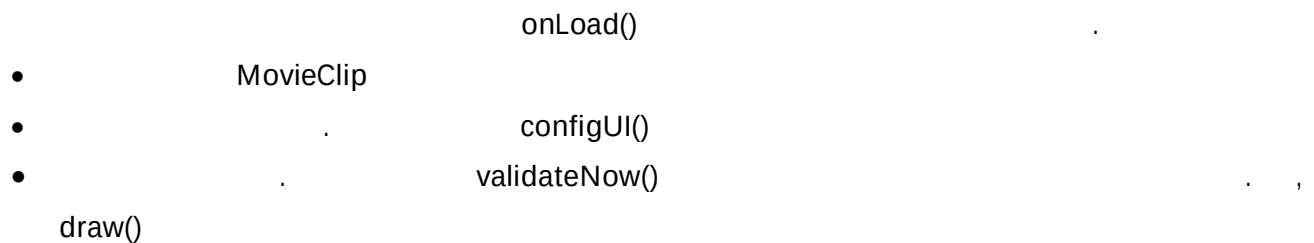
4.1 UIComponent

CLIK

UIComponent(gfx.core.UIComponent)



4.1.1 초기화



4.2

CLIK

4.2.1 Button 컴포넌트

(
CLIK
ListItemRenderer,
RadioButton, CheckBox
ScrollBar
Button
CLIK
Button
(gfx.controls.Button)
Button

- *up*
- *over*
- *down*
- *disabled* disable

Button (prefixed state)
CLIK Button “selected_”

(selected) Button

- *up*
- *over*
- *down*
- *disabled* disable
- *selected_up*
- *selected_over*
- *selected_down*
- *selected_disabled*

Getting Started with CLIK Buttons

```
private function getStatePrefixes():Array {
    return (_selected) ? ["selected_", ""] : [""];
}
```

```

    , (
    . Button stateMap . CLIK

```

```
private var stateMap:Object = {
  up:["up"],
  over:["over"],
  down:["down"],
  release: ["release", "over"],
  out:["out", "up"],
  disabled:["disabled"],
  selecting: ["selecting", "over"],
  kb_selecting: ["kb_selecting", "up"],
  kb_release: ["kb_release", "out", "up"],
  kb_down: ["kb_down", "down"]
}
```

97

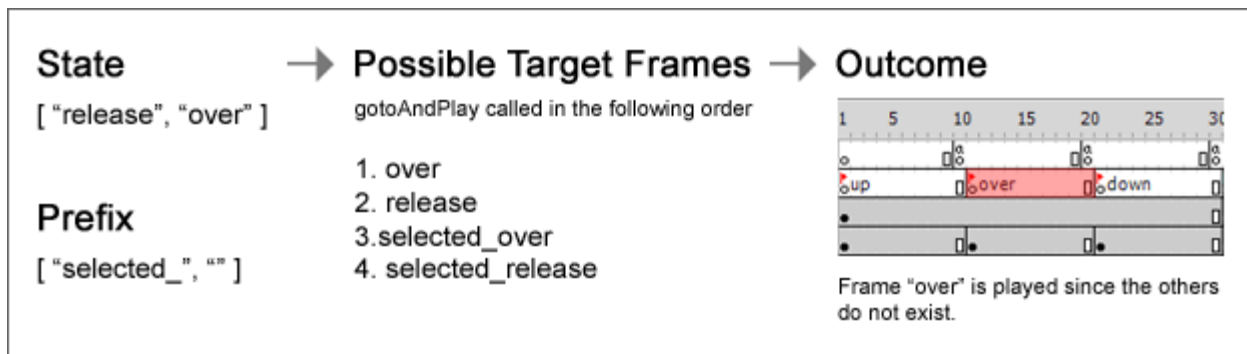


그림 61: 올바른 재생 키프레임을 결정하는데 사용되는 프로세스

4.2.2 버튼이 아닌 상호작용 컴포넌트

CLIK

ScrollingList, OptionStepper, Slider, TextArea

- *default*
- *focused*
- *disabled*

4.2.3 비 상호작용 컴포넌트

, disable

- *default*
- *disabled*

4.2.4 특수한 경우

. StatusIndicator

ProgressBar

(%)

0, 10 50

```

        StatusIndicator , 5(50%) gotoAndStop()
25(50 50%)
        .updateValue()
        .
    ,
        . TextInput TextArea
asAsButton roll over roll out over
out
    .

```

4.3

CLIK

“ (dispatch)” . ,

```

EventDispatcher(gfx.events.EventDispatcher) API
    . CLIK EventDispatcher.initialize()
        CLIK UIComponent super()
        UIComponent EventDispatcher
EventDispatcher

```

4.3.1 사용예와 가장 좋은 연습(Usage and Best Practices)

4.3.1.1 이벤트에 신청하기(Subscribing to an Event)

```

        addEventListener()
        removeEventListener()

```

EventTypes(gfx.events.EventTypes)

(EventTypes "show" SHOW).

```
buttonInstance.addEventListener("itemClick", this, "callBack");
function callBack(eventObj:Object):Void {
    buttonInstance.removeEventListener("itemClick", this, "callBack");
}
```

buttonInstance itemClick
callback callback buttonInstance
CLIK
([])

4.3.1.2 이벤트 발송(Dispatching an Event)

CLIK

dispatchEvent()

```
dispatchEvent({type:"itemClick", item:selectedItem});
```

4.4

CLIK ()
,
Scaleform ()
Selection.setFocus(instance) . FocusHandler
(gfx.managers.FocusHandler)
FocusHandler

4.4.1 사용예와 가장 좋은 연습(Usage and Best Practices)

```

        CLIK
        ).
        ( , ).
true
    .

```

```
Selection.setFocus(firstComponentOrMovieClip);
```

```

        ( : onRelease, onRollOver)      MovieClips      Scaleform
        .
        . Button
        MovieClip focusEnabled
tabEnabled
    .

```

```

        (ScrollBar )      focusTarget
        .      FocusHandler
        focusTarget      focusTarget
    .

```

```

    . displayFocus      true
    Slider      Slider
        UIComponent.changeFocus()
    .

```

```

function changeFocus():Void {
    track.displayFocus = _focused;
}

```

```

        .
        tabEnabled
false
    .

```

```
background.tabEnabled = false;
```

4.4.2 복합 컴포넌트에서 포커스 캡처

```

OptionStepper, ButtonBar
, ScrollingList,
Scaleform
1. tabEnabled = false ( , OptionStepper
2. focusTarget
3. focusEnabled = true
changeFocus displayFocus true

```

4.5

CLIK

8 MovieClip

MovieClip

.

.

, CLIK

.

4.5.1 사용예와 가장 좋은 연습(Usage and Best Practices)

```
graph TD; InputDelegate[InputDelegate] --> InputDetails[InputDetails]; InputDetails --> InputDelegate;
```

- “key”
-
- . Key key up key down
, InputDetails value “KeyUp”
“KeyDown”
- “up”, “down”, “left”, “right”
navEquivalent (navigation equivalent)
NavigationCode(gfx.ui.NavigationCode)

FocusHandler

InputDelegate

handleInput()

, InputDetails pathToFocus

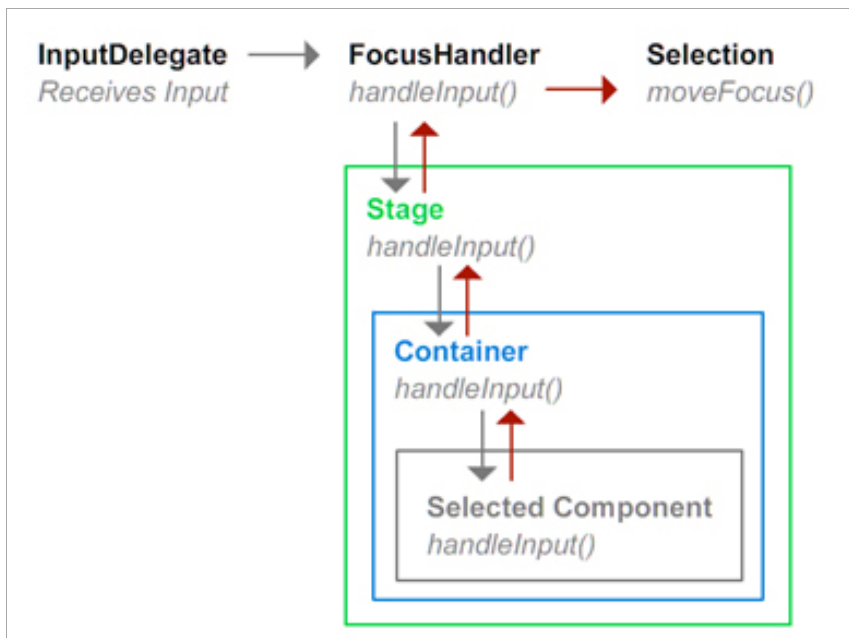


그림 62: handleInput() 체인

FocusHandler handleInput()

. false , navEquivalent null

Scaleform

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    if (details.navEquivalent == "left")
    { // or NavigationCode.LEFT
        doSomething();
        return true;
    }
    return false;
}
```

pathToFocus

true false

handleInput

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var nextItem:MovieClip = pathToFocus.shift();
    var handled:Boolean = nextItem.handleInput(details, pathToFocus);
    if (handled) { return true; }
    // custom handling code
    return false; // or true if handled
}
```

DropDownMenu

handleInput dropdown

pathToFocus

```
function handleInput(details:InputDetails, pathToFocus:Array):Boolean {
    var handled:Boolean = dropdown.handleInput(details);
    if (handled) { return true; }
    // custom handling code
    return false; // or true if handled
}
```

InputDelegate.instance

FocusHandler

```
InputDelegate.instance.addEventListener("input", this, "handleInput");
function handleInput(event:Object):Void {
```

```

        var details:InputDetails = event.details;
        if (details.value == Key.TAB) { doSomething(); }
    }

```

InputDelegate

InputDelegate

W, A, S, D

4.5.2 다중 마우스 커서

(Wii™)

. CLIK

SWF

mouseIndex

, mouseIndex

```

myButton.addEventListener("click", this, "onCursorClick");
function onCursorClick(event:Object):Void {
    switch (event.mouseIndex) {
        ...
    }
}

```

4.6

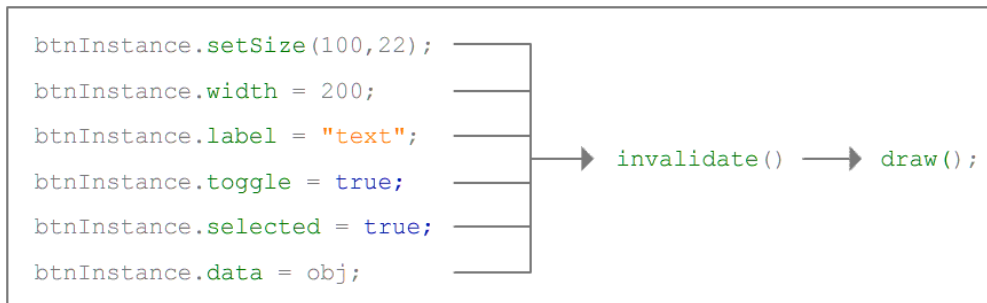


그림 63: CLIK 컴포넌트는 특정 속성이 변경되면 자동적으로 무효화한다.

4.6.1 사용예와 가장 좋은 연습(Usage and Best Practices)

set
), UIComponent.invalidate()
UIComponent.draw()
invalidate()
draw()
UIComponent.validateNow()

4.7

CLIK 2

1. reflowing

2. (counterscaling)

, scale9grid,

Flash scale9Grid reflow
"skin"

. Reflowing
container

Flash/Flex

Scaleform

scale9Grid

CLIK
layered

CLIK

setups(

...)

container

Reflow

IDE

, setSize()

, width,

height

IDE

MovieClips

LivePreview

Scaleform Player

. CLIK

4.7.1 Scale9Grid

. Scaleform Player

Scale9Grid

MovieClip

scale9grid

,

MovieClip

scale9grid

Scaleform

MovieClip scale9grid

. scale9grid

4.7.2 Constraints

Constraints(gfx.utils.Constraints)

, ScrollBar

. Constraints

```

        ScrollBar
        ,
        configUI()
        ,
        constraints
        ,
        . draw()
        ,
        constraints
        ,
        draw()
        ,
        (
        ,
        ).

```

```

private function configUI():Void {
    ...
    constraints = new Constraints(this);
    // The upArrow already sticks to top left.
    constraints.addElements(downArrow, Constraints.BOTTOM);
    constraints.addElenent(track, Constraints.TOP|Constraints.BOTTOM);
    ...
}

private function draw():Void {
    ...
    constraints.update(__width, __height);
    ...
}

```

4.8 *(Components and Data Sets)*

```

        dataProvider
        , IDataProvider(gfx.interfaces.IDataProvider)
        , dataProvider
        , API
        ,
        dataProvider
        ,
        dataProvider
        ,
        dataProvider
        , CoreList (ScrollingList, TileList), OptionStepper,
        DropdownMenu
        , CLIK
        , IDataProvider
        , API
        , IDataProvider
        ,

```

4.8.1 **사용예와 가장 좋은 연습(Usage and Best Practices)**

```

        DataProvider (gfx.data.DataProvider)
        , initialize()
        , dataProvider
        , dataProvider
        , dataProvider
        ,

```

, IDDataProvider dataProvider

```
myComponent.dataProvider = [ "data1",
                              4.3,
                              {label:"anObjectElement", value:6} ];
```

dataProvider

- *length*: get
- *requestItemAt*: dataProvider OptionStepper
- *requestItemRange*: dataProvider ScrollingList
- *indexOf*:
- *invalidate*: dataProvider

“dataChange” dataProvider
length

dataProvider

2

dataProvider ExternalInterface.call

4.9

CLIK Tween (gfx.motion.Tween) 8

Tween , Scaleform

Tween mx.transitions.easing.* package easing

, CLIK Tween , tween 8

MovieClip tween CLIK

MovieClip CLIK Tween onEnterFrame

MovieClip tween tween

```

tween      MovieClip      tween      tween
    tween      MovieClip
        tweening      tween

```

4.9.1 사용예와 가장 좋은 연습(Usage and Best Practices)

```

Tween      2      MovieClip: tweenTo()      tweenFrom()
tweenTo()      MovieClip      tweening
tweenFrom()      tweening

```

```

tween      MovieClip      Tween
        Tween.init()

```

```

import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init(); // Install tween methods to MovieClip prototype
// Perform a 1 second tween from the current horizontal position
and
    alpha values to the ones specified
myMovieClip.tweenTo(1, {_x: 200, _alpha: 0}, Strong.easeIn);

```

```

tween      onTweenComplete      tween

```

```

import mx.transitions.easing.*;
import gfx.motion.Tween;
Tween.init();
mc.onTweenComplete = function() {
    trace("Tween has finished!");
}
mc.tweenTo(5, {_rotation: 20}, Bounce.easeOut);

```

4.10

```

CLIK      PopUpManager (gfx.managers.PopUpManager)
        Dialog      PopUpManager

```

4.10.1 사용예와 가장 좋은 연습(Usage and Best Practices)

```

PopUpManager
createPopUp()      MovieClip      ID

```

(CLIK
relativeTo
). Context

```
import gfx.managers.PopUpManager;  
PopUpManager.createPopUp("context, "MyToolTipLinkageID",  
{_x: 200, _y:200}, relativeTo);
```

Scaleform
topmostLevel
root
SWF
SWF
context
topmostLevel
topmostLevel
ordering
topmostLevel
true
z-

4.11

DragManager (gfx.managers.DragManager)
(singleton)

startDrag() MovieClipd ID
stopDrag() DragManager
InputDelegate

4.11.1 사용예와 가장 좋은 연습(Usage and Best Practices)

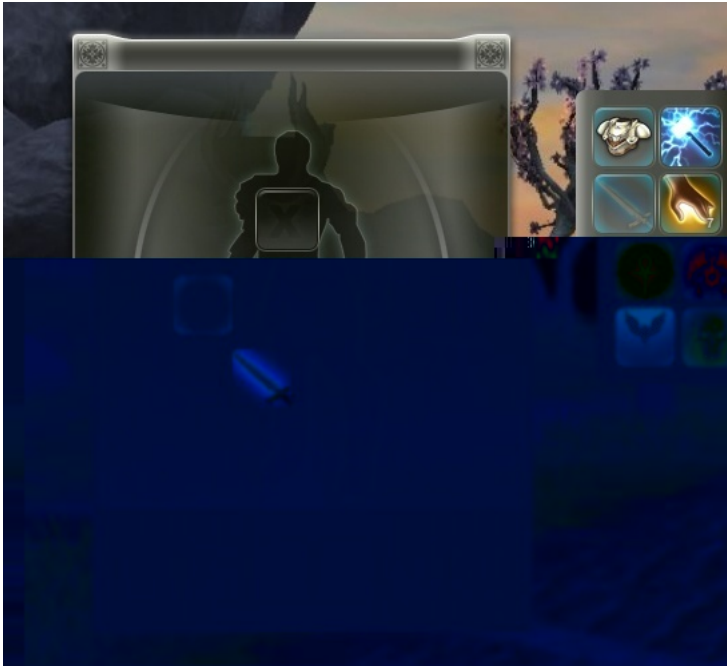


그림 64: 작동중인 드래그 데모 DragDemo

```

DragManager . &
DragManager . CLIK
DragTarget (gfx.controls.DragTarget) .
DragTarget UIComponent CLIK ,
DragManager .

DragTarget . DragTarget
DragTarget DragManager
dragBegin dragEnd . DragTarget
DragTarget

CLIK Inventory DragTarget 2
DragManager DragTarget &
CLIK ,

```

```

InventorySlot(gfx.controls.InventorySlot)      RPG
                                                DragTarget
                                                .
                                                , DragManager
                                                .

InventorySlot
.

ItemSlot (gfx.controls.ItemSlot)      InventorySlot
. ItemSlot      DragTarget
.      2
.      , CLIK Button
ItemSlot      &

```

4.12 (Miscellaneous)

4.12.1 Delegate

```

Delegate (gfx.utils.Delegate)
.      CLIK      ,      DragManager
.

```

```

eventProvider.onMouseMove = Delegate.create(this, doDrag);

```

4.12.2 로케일(Locale)

```

Locale (gfx.utils.Locale)      Localization      . Scaleform
Localization
.      ,      . Button, Label,
TextInput      Locale.getTranslatedString()
Localization
Locale      Localization
.

```


5 사용예

CLIK

5.1

CLIK

5.1.1 2 개의 textField 를 가진 ListItem 렌더러

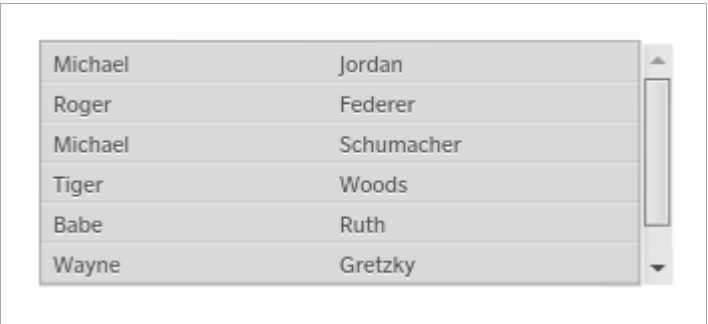
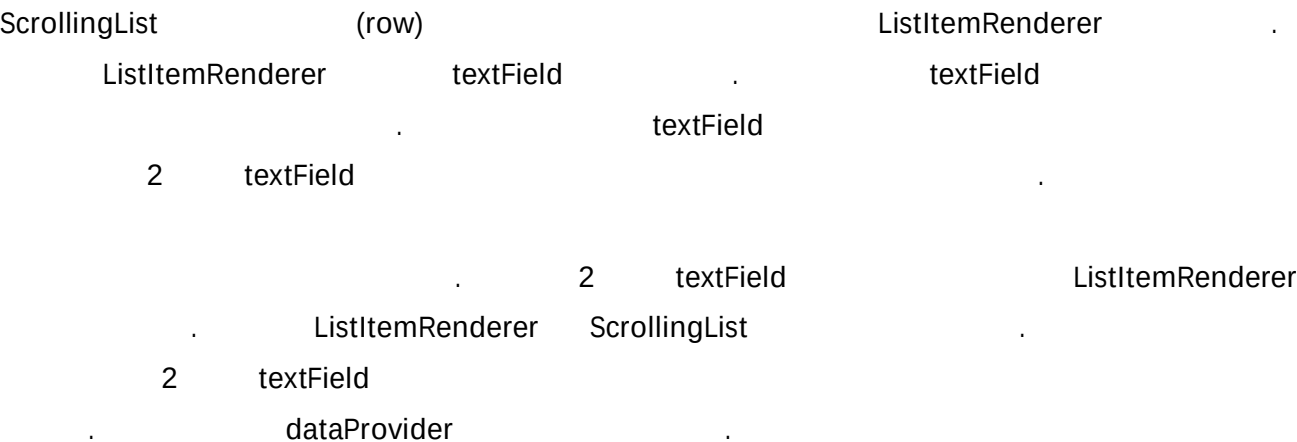


그림 65: 2 개의 레이블을 포함한 리스트 항목을 보여주는 ScrollingList



```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger", lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger", lname: "Woods"},
                     {fname: "Babe", lname: "Ruth"}]
```

```
{fname: "Wayne", lname: "Gretzky"},
{fname: "Usain", lname: "Bolt"}];
```

```
2 (fname lname) 2
textField 2
ListItemRenderer textField , 2 textField
ListItemRenderer
MyItemRenderer , ListItemRenderer 2
textField . ( FLA
MyItemRenderer.as )
```

```
import gfx.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

    public var textField1:TextField;
    public var textField2:TextField;

    public function MyItemRenderer() { super(); }

    public function setData(data:Object):Void {
        this.data = data;
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }

    private function updateAfterStateChange():Void {
        textField1.text = data ? data.fname : "";
        textField2.text = data ? data.lname : "";
    }
}
```

```
ListItemRenderer setData updateAfterStateChange
(override) . setData
(ScrollingList, TileList, ) . ListItemRenderer
textField , MyItemRenderer
textField ,
updateAfterStateChange , ListItemRenderer
textField
```

```

dataProvider
ListItemRenderer
    .
    ListItemRenderer
    ListItemRenderer
    'textField1' 'textField2' 2 textField
    ID
    MyListItemRenderer
    . MyListItemRenderer
itemRenderer 'ListItemRenderer' 'MyListItemRenderer'
    .

FLA . 2 . fname
lname dataProvider
    .

```

5.1.2 픽셀별 스크롤 뷰

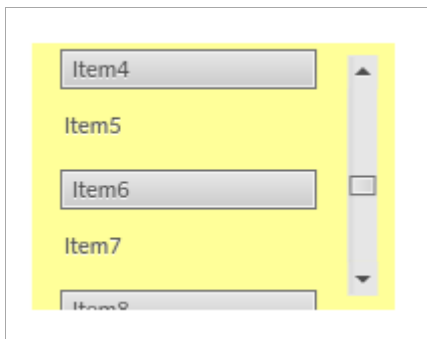
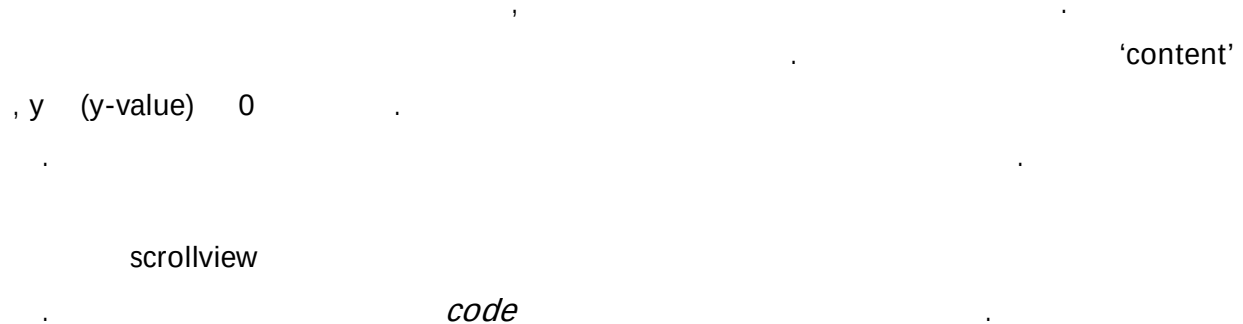


그림 66: CLIK 요소를 컨텐트로 포함하고 있는 픽셀별 스크롤 뷰

fn/C20*13Tf20h1 Tf2.489 0 Td4.7009F111009B7215 0 Td02A8-110887

- *background:*



```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content._height - 133);
sb.position = 0;

sb.addEventListener("scroll", this, "onScroll");
function onScroll() {
    content._y = -sb.position;
}
```

Scrollbar

setScrollProperties

, scrollbar

ScrollView scrollbar

Scaleform

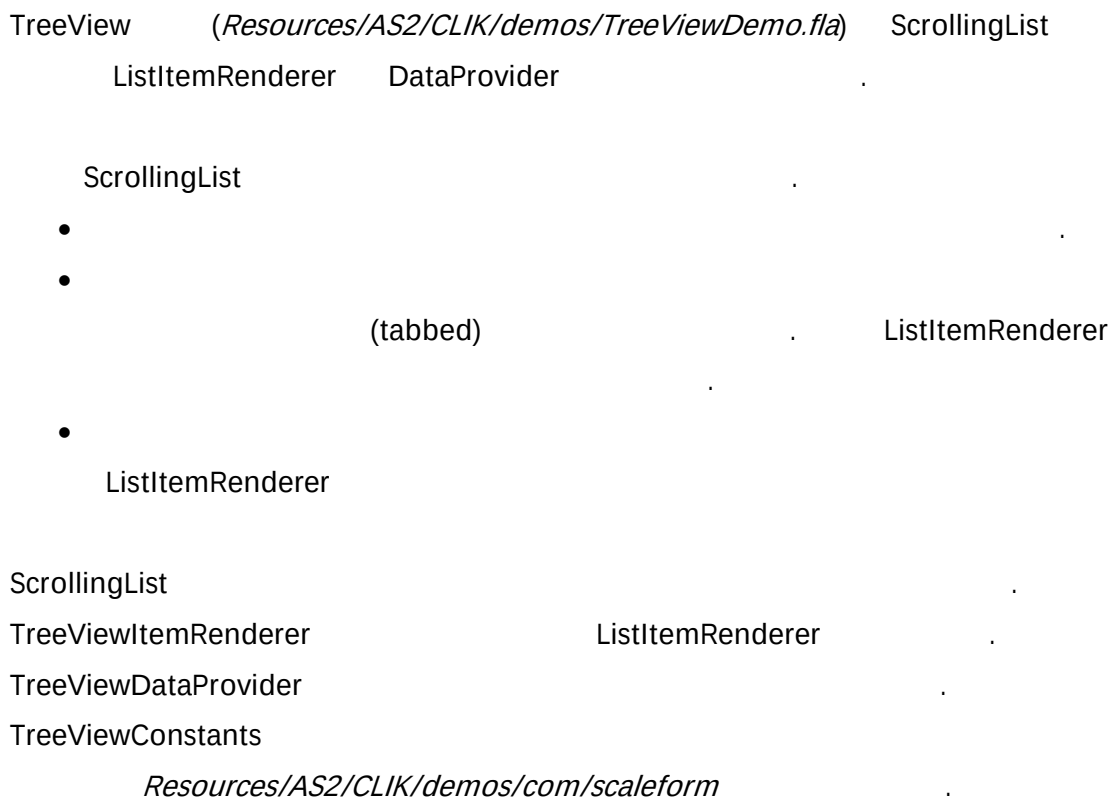
HUD

5.2 (Complex)

CLIK

5.2.1 ScrollingList 를 사용중인 TreeView

그림 67: TreeView 데모.



```

var root:Object =
{
    label: "System",
    nodes:[
        {label: "Applications",
        nodes:[
            {label: "CGStudio",
            nodes:[
                {label: "Data"},
                {label: "Samples"},
                {label: "Config"}
            ]
            },
            {label: "Money Manager"},
            {label: "Role Call v6"},
            {label: "Inventionista"},
            {label: "Super Draft",
            nodes:[
                {label: "Templates"},
                ...
            ]
            }
        ]
    ]
}

```

2 (label nodes) ScrollingList

TreeViewDataProvider

ListltemRenderer

. TreeViewDataProvider

CLIK

TreeViewltemRenderer ScrollingList TreeViewDataProvider

ListltemRenderer

TreeViewltemRenderer

'tab'

```

        . ListItemRenderer textField
        .
    }

    TreeViewConstants
    {
        •
        •
        •
    }

    /

    ScrollingList ScrollBar
    /

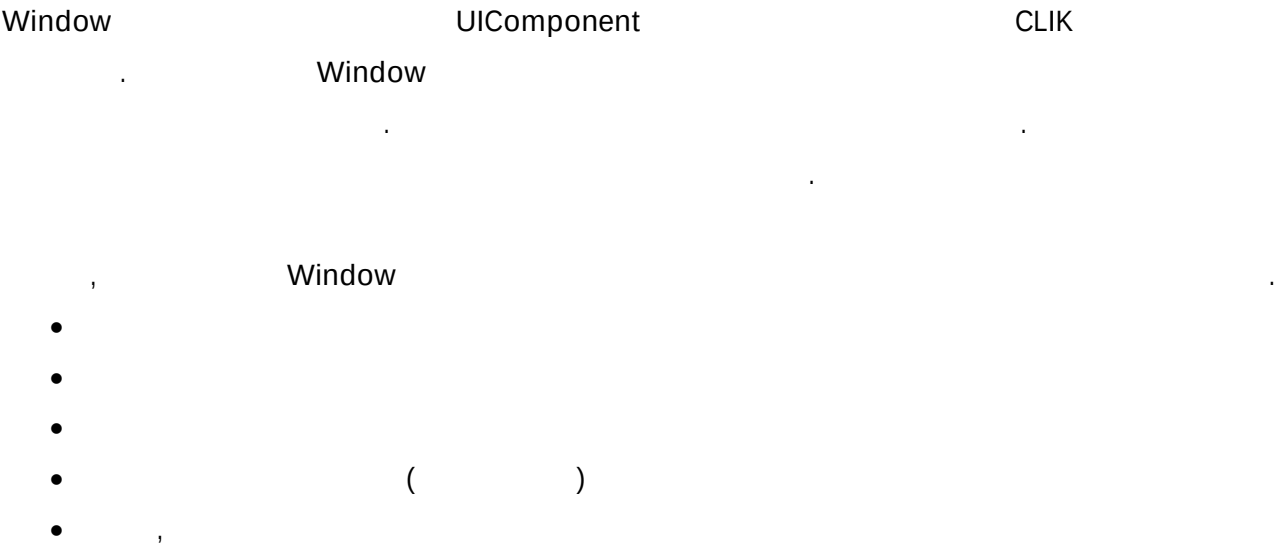
    (grip)

```

5.2.2 재사용 가능한 윈도우

그림 68: Window 데모.

Window (*Resources/AS2/CLIK/demos/WindowDemo.fla*)
 . Window
Resources/AS2/CLIK/demos/com/scaleform/



Window

Title	Window
formPadding	Window
formType	'symbol' 'swf' 'symbol' 'swf' SWF
formSource	(formType 'symbol') (formType 'swf')SWF
allowResize	
minWidth,	. max min
maxWidth,	. max
minHeight,	. max 0
maxHeight	

get

set

```
[Inspectable(name="formType", enumeration="symbol, swf")
private var _formType:String = "symbol";
```


formType

formType

set

get

set, get

```
[Inspectable(defaultValue="Title")]
public function get title():String { return _title; }
public function set title(value:String):Void {
    _title = value;
    invalidate();
}
```

Window

CLIK Button

Window

UIComponent

2

config)

draw5D7-30))TJ/C2_0 1 Tf0 Tc 0 Tw 2.815 0 Td[0571517F]TJ/TT0 1 Tf()Tj/C2_0 1 Tf1.5 0 Td[08A211090F08BD0B1E042C]TJ/TT0 1 Tf0 Tc 0 Tw 2.815 0 Td[05A907150496110415]TJ/TT0 1 Tf()Tj/C2_0 1 Tf-0.011 Tc 0.011 Tw 4.489 0 Td[088A>11060608B

constraints

draw()

cnstraints

vaidaeNo()

Component

dra()

valideNow()

draw()

UIComponent

Window
SWF

```
draw()    validateNow()
```

Window

Window

```
private function onMouseDown() {
    var targetObj:Object = Mouse.getTopMostEntity();
    while (targetObj != null && targetObj != _root)
    {
        if (targetObj == this) {
            swapDepths(_parent.getNextHighestDepth());
            return;
        }
        targetObj = targetObj._parent;
    }
}
```

WindowManager

$$Z$$

. WindowManager